

Few-Shot Prediction of Post-Fusion Acute Kidney Injury Using Multimodal Radiomics-Clinical Feature-Based Prototype Networks

--Manuscript Draft--

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Full Title:	Few-Shot Prediction of Post-Fusion Acute Kidney Injury Using Multimodal Radiomics-Clinical Feature-Based Prototype Networks
Short Title:	Few-Shot Prediction of Post-Fusion Acute Kidney Injury
Corresponding Author:	Zhen Yang Chengdu Second People's Hospital CHINA
Keywords:	Acute kidney injury (AKI) , Few-shot learning , Prototypical networks , Multimodal fusion , SHAP value analysis
Abstract:	To optimize preoperative decision-making, the prediction of acute kidney injury (AKI)—a major postoperative complication—requires early identification. This study developed a multimodal few-shot prototype network based on real-world clinical data from 194 single-center cases. The model incorporated 11 preoperative parameters encoded by a multilayer perceptron (MLP) to weight and integrate imaging features extracted by ResNet-50, with a fixed loss function implemented to address class imbalance. Results demonstrated that the fused model achieved an area under the curve (AUC) of 0.816, significantly outperforming the clinical-only (AUC = 0.578) and imaging-only (AUC = 0.510) models. Interpretability analysis using SHapley Additive exPlanations (SHAP) revealed the influence of key variables (age, lactate, creatinine). Our study demonstrates that multimodal fusion significantly improves generalization performance in data-limited scenarios, establishing the first application of few-shot learning with prototype networks for postoperative AKI prediction. We recommend incorporating 11 standardized preoperative indicators and imaging data into clinical risk assessment to enhance predictive accuracy and utility. Future directions should focus on expanding cohort sizes and refining radiomic feature extraction.
Order of Authors:	JingLan Zhang Zhen Yang Jun Li
Opposed Reviewers:	
Response to Reviewers:	Manuscript title: Few-Shot Prediction of Post-Fusion Acute Kidney Injury Using Multimodal Radiomics-Clinical Feature-Based Prototype Networks Journal: PLOS ONE Jing Lan Zhang¹ College of Physics and Electronic Engineering, Sichuan Normal University, Chengdu, 610101, China. * Correspondence: Corresponding Author: ZhenYang , Jun Li yangzhen1219@outlook.com lijun2022@sicnu.edu.cn Manuscript ID: PONE-D-25-38340 Dear Dr. Irfan Ullah and Reviewers, Thank you for the constructive feedback. We have revised the manuscript accordingly. In addition to the text below, we have uploaded an identical .docx file titled "Response to Reviewers.docx." Below, we provide a point-by-point response—each item begins with Comment followed by Response—and we indicate the exact locations of the corresponding changes in the Revised Manuscript (with Track Changes) and the Clean Manuscript (page and line numbers). Replacement text for substantial rewrites is provided in the Appendix.

Reviewer #1 — 1. Page 12, Lines 251–258 (Manuscript length & repetition; KDIGO definition duplicated)

Comment: “The manuscript is overly long and contains considerable repetition with unclear structure. For example, the KDIGO definition of AKI appears verbatim in both the Introduction and Methodology. Please streamline the text, remove redundancies, and focus on the most relevant content.”

Response: Thank you for this important comment. We have substantially streamlined and restructured the relevant sections and have removed the verbatim KDIGO definition from this (Methodology) passage. The formal KDIGO definition now appears only once in the Introduction with citation. In the Methodology, we no longer repeat the definition; instead, we include a one-sentence cross-reference to the Introduction and retain only the baseline SCr selection rule necessary for outcome adjudication, thereby avoiding redundancy while ensuring reproducibility and auditability.

Actions taken:

Remove the duplicate definition: Deleted the verbatim KDIGO definition from Methodology; retained a single, authoritative KDIGO definition only in the Introduction (with references).

Retain essential rule and add cross-reference: In Methodology (Outcome definition), added a one-sentence note—“AKI definition, see Introduction”—and kept only the baseline SCr selection rule to support outcome adjudication; KDIGO thresholds and time windows are not repeated.

Consolidate redundant statements: Merged statements regarding “retrospective design, anonymization, and lack of access to identifiable information” into a single paragraph (see replacement text below), and centralized ethics and data-availability details in their dedicated sections/Supplement to avoid scatter.

Location in the revised manuscript: (Manuscript) Introduction—Background/Definitions (the KDIGO formal definition appears only once): page 3, lines 45–48.

Study population / Data sources (revised paragraph replacing the prior KDIGO duplication): page 12, lines 251–258.

Reviewer #1 — 2. Page 6, Lines 115–131 (Rationale for focusing on AKI after laparoscopic renal surgery)

Comment: “The rationale for focusing on AKI after laparoscopic renal surgery (partial resection/nephrectomy) is underdeveloped. The manuscript does not explain why this surgical context was chosen or why a dedicated prediction model is needed. Please clarify in the Introduction why this setting is a priority for investigation.”

Response: We appreciate this important suggestion and agree that the original manuscript did not sufficiently articulate the study rationale. In accordance with your advice, we have substantively rewritten and expanded the relevant paragraph in the Introduction to:

- (1) explicitly state that laparoscopic nephrectomy involves dual renal stress mechanisms—pneumoperitoneum-induced reductions in renal perfusion and ischemia–reperfusion injury during hilar clamping—thereby explaining, on a pathophysiological basis, why this setting is more prone to AKI;
- (2) add the incidence range (15%–25%) under KDIGO criteria and the robust association between postoperative AKI and long-term renal function decline/new-onset CKD, underscoring clinical importance;
- (3) clarify that generic perioperative AKI models, mostly derived from cardiac or heterogeneous surgical cohorts, omit procedure-specific variables (e.g., pneumoperitoneum pressure, warm ischemia time), limiting transportability and predictive performance in this setting;
- (4) consequently, highlight the need for a surgery-specific prediction model tailored to this context to address the research gap and clinical demand. All assertions are supported with targeted references.

Actions taken:

Replaced the original lines 115–118 with a structured rationale covering four layers—mechanism, epidemiology, prognosis, and model gap—and added/updated references to support each key point.

Ensured consistency by introducing and defining procedure-specific variables (pneumoperitoneum pressure, warm ischemia time) in the subsequent Methods and variable lists, thereby closing the loop with the Introduction.

Location in the revised manuscript: (Manuscript) Introduction, page 6, lines 115–131

Reviewer #1 — 3. Pages 18–44 (Methods overly long; details should be moved to Supplement)

Comment: “The Methods section is excessively long (approximately 20 pages) and is not appropriate for the main manuscript. Only essential methodological details should be retained, while detailed definitions and extended technical explanations should be moved to the Supplement to improve readability and focus.”

Response: We appreciate this key suggestion. We agree that the main-text Methods should focus on core workflow and decisions, while implementation details, parameters, and extended experiments should be placed in the Supplement. Accordingly, we performed a structural condensation and rewrite of the Methods, retaining only the key elements needed for reproducibility in the main text with pointer citations, and moving all additional technical details to the Supplement.

1.Modification 1 (page 18, lines 369–392)

Actions taken:

Merge and refocus: We merged the original 3.2 Data Preprocessing and 3.3 Data Augmentation into 3.2 Data Preprocessing and Augmentation, retaining only:

Clinical branch: median imputation and Z-score normalization ;

Imaging branch: YOLO-based ROI cropping and uniform resolution (with a pointer to Supporting information S2);

Class imbalance: feature-space interpolation & weighted sampling & class-weighted loss within the multimodal framework;

Anchor-based pseudo-sample strategy presented as a high-level overview (linked to Fig. 2), with algorithmic details moved to Supporting information S2.

Details moved to the Supporting information:

Supporting information S1 (Data Imputation and Standardization): implementation details for imputation/standardization and train–inference consistency;

Supporting information S2 (Data Augmentation, Class Balancing, and Feature Extraction): augmentation parameters (e.g., $\pm 5^\circ$ rotation, $\pm 5\%$ brightness/contrast, 224×224), backbone and feature dimensions (e.g., ResNet-50 without classification head, 2048-D), SMOTE-inspired interpolation, WeightedRandomSampler & class-weighted loss, and the three-step anchor-based generation with thresholding.

Remove redundancy and engineering minutiae: Eliminated repeated descriptions of standardization, duplicate phrasing, library/function names, and serialization steps from the main text; replaced with concise pointers “see Supporting information S1/S2.”

Length reduction: The main-text Methods are substantially shorter than in the original submission (redundancies and implementation details removed), yielding a tighter, clearer presentation that highlights study design and primary innovations.

Location in the revised manuscript:

Methodology — 3.2 Data Preprocessing and Augmentation, page 18, lines 369–392; Fig. 2 (anchor-based strategy schematic).

Supporting information S1: Data Imputation and Standardization; Supporting information S2: Data Augmentation, Class Balancing, and Feature Extraction.

2.Modification 2 (page 19, lines 393–408)

Actions taken:

Pruned & merged: Deleted the former 3.4.1 Design Rationale and Architecture and merged its key points into the opening paragraph of the fusion section (now 3.3).

Retained essentials only: Kept a high-level description of the architecture and key steps to avoid overlap with figures/other subsections.

Location in the revised manuscript:

Methodology — 3.3 Multimodal Feature Fusion Network (one-paragraph summary after merging), page 19, lines 393–408.

3.Modification 3 (page 20, lines 410–418)

Actions taken:

Pruned and merged: Deleted content in the original 3.4.2 that duplicated the overview (e.g., background suppression from ROI cropping, 2048-D GAP outputs, flexible fine-tuning, and inference efficiency) and rewrote the subsection as a concise 3.3.1 Feature Extraction Module.

Retained essential points only: In the main text, we keep two high-level items: Image feature encoder: pretrained ResNet-50 with YOLO-based ROI cropping to obtain high-level visual features; Tabular feature encoder: shallow MLP + ReLU + Dropout to

produce a compact structured representation; and add a one-sentence pointer to the Supporting information for implementation details.

Moved details to the Supporting information: Transferred network architecture, parameter settings, and training strategy to Supporting information S3 (S3.1–S3.3), including ResNet-50 without the classification head + GAP (2048-D), ROI pixelization and resizing to 224×224, a two-layer MLP (ReLU / Dropout = 0.2), and output dimensions (image 2048, tabular 128).

Location in the revised manuscript:

Methodology — 3.3.1 Feature Extraction Module, page 20, lines 407–418 (concise description with a terminal pointer to Supporting information S3).

Supporting information S3 (S3.1 Image Feature Encoder; S3.2 Tabular Feature Encoder; S3.3 Module Parameters and Compatibility).

4.Modification 4 (page 20, lines 419–434)

We merged the former 3.4.3 (Dynamic Gated Fusion Mechanism) and 3.4.4 (Multimodal Feature Fusion and Dimensionality Reduction) into the new 3.3.2 Dynamic Gated Fusion and Dimensionality Reduction. In the main text, we now provide a single, high-level paragraph describing the fusion logic and dimensionality-reduction flow, while formulas and parameter details are moved to the Supporting information, improving readability and focus.

Actions taken:

Pruned & merged: Removed repetitive/stepwise exposition across the two subsections and integrated the pipeline—sample-wise sigmoid gating → weighted reconstruction → lightweight two-layer projection to 128-D → L2 normalization → prototype mean computation—into a coherent description under the new 3.3.2.

Essential information retained: We state that sample-wise sigmoid gating replaces softmax competition to jointly amplify informative modalities and down-weight unreliable inputs; the gated vector is projected via two lightweight layers to a 128-D embedding, then L2-normalized to the unit sphere to ensure metric consistency and interpretability with Prototypical Networks.

Details moved to the Supplement: Formal derivations, network parameters, and implementation details are now in Supporting information S4 (Dynamic Gated Fusion Mechanism) and Supporting information S5 (Multimodal Feature Fusion and Dimensionality Reduction), avoiding length in the main text.

No extra hyperparameter tuning: We clarify that this design improves sensitivity and generalization under small-sample and class-imbalanced conditions without introducing additional hyperparameter tuning.

Location in the revised manuscript:

Methodology — 3.3.2 Dynamic Gated Fusion and Dimensionality Reduction, pages 20–21, lines 419–434.

Supporting information S4 (Dynamic Gated Fusion Mechanism); Supporting information S5 (Multimodal Feature Fusion and Dimensionality Reduction).

5.Modification 5 (page 21, lines 436–452)

Actions taken:

Revised and condensed the original 3.5.1 Multi-Prototype Construction and moved implementation details to Supporting information S6 (now presented as 3.4.1 Multi-Prototype Construction in the main text).

Key points retained : We note that a single prototype is insufficient when there is intra-class multimodality/sub-category structure; we therefore adopt a KMeans-based multi-prototype scheme on support-set embeddings, where multiple prototypes jointly describe each class distribution. For classes with very few samples, we apply a mean-based compensation to maintain a fixed number of prototypes. We also cite prior work showing the benefits of multi-prototype representations in fine-grained / high intra-class variability settings.

Details moved to the Supplement: The clustering objective, choice of number of prototypes, and implementation/training interface are all placed in Supporting information S6 (including formulas and implementation notes in S6.1). The main text points readers to “see Supporting information S6” and omits derivations and parameters.

Tightened exposition: The main text emphasizes that this clustering step is an offline, unsupervised aid that does not interfere with the supervised training objective, while enriching class structure and improving the stability and accuracy of metric-based inference.

Location in the revised manuscript: Methodology — 3.4.1 Multi-Prototype Construction, page 21, lines 436–452. Supporting information S6 (Few-Shot Learning Strategy), S6.1 Multi-Prototype Construction Details (including formulas and implementation details). 6.Modification 6 (page 22, lines 453–466) Actions taken: Core logic retained: We state that all prototypes are L2-normalized to eliminate scale differences. During classification, similarity to a class is computed as the minimum Euclidean distance to that class’s prototypes, aligning each query with the nearest local pattern within the class and thereby improving accuracy and robustness. Location in the revised manuscript: Methodology — 3.4.2 Prototype Normalization and Minimum-Distance Selection, page 21, lines 453–466. Supporting information S6.2 (Prototype Normalization and Minimum-Distance Selection; formulas and implementation details). 7.Modification 7 (page 24, lines 482–501) Actions taken: Moved details to the Supplement: We transferred the detailed derivations, parameter settings, and ablation studies from the original 3.6.1–3.6.4 to Supporting information S7. In the main text, we now provide a single-paragraph overview under 3.5 Loss Function Design, clarifying the roles and rationale of the three losses and their final weighting to improve readability and focus. Overview retained in the main text (functions and roles): 3.5 Loss Function Design—Multi-prototype classification loss: captures multi-peak/multimodal intra-class structure, compensating for the limitations of single-center prototypical networks; Anchor alignment loss: enforces semantic consistency between pseudo-samples and class anchors (e.g., SwAV vectors), suppressing semantic drift and improving pseudo-sample reliability; Batch-hard triplet loss: focuses on the most discriminative positive–negative pairs, tightening inter-class boundaries, accelerating convergence, and improving separability; Weighted fusion: combines the sub-losses into a unified objective to balance accuracy, semantic alignment, and embedding-space discriminability. Details deferred to the Supplement: Mathematical formulations, parameter configurations, and ablation analyses for each loss are presented in S7. The main text references these details via “see Supporting information S7” and does not repeat derivations or engineering specifics. Streamlining and consistency: We removed technical statements that duplicated other sections, and unified terminology and citation ranges. Loss weights and sensitivity analyses are reported in S7, aligned with the Results discussion. Location in the revised manuscript: Methodology — 3.5 Loss Function Design, page 24, lines 482–501. Supporting information 7 (formulas, parameter settings, and ablation studies for all losses). Reviewer #1 — 4. Page33 lines 669-678. (Temporal validity not discussed) Comment: “Temporal validity should be explicitly discussed; the model needs periodic retraining and external validation on newer data, otherwise performance may degrade.” Response: Thank you for highlighting this important limitation. We have added a new “Temporal Validity” paragraph to the Discussion—Limitations section. We explicitly acknowledge that a model trained on historical single-center data may experience performance drift over time. We also outline mitigation strategies, including temporal external and multicenter validation, routine performance monitoring and recalibration, and integration with hospital information systems to ensure stability and applicability in real-world clinical settings. In addition, we describe future work on dynamic, continuously updated prediction frameworks and clarify current limitations in generalizing to specific populations (e.g., patients with a single kidney, severe renal impairment, or children), together with planned improvements. Actions taken: Added a dedicated subsection after Limitations discussing temporal validity and the risk of performance degradation as clinical practice evolves and data distributions shift. Committed to temporal external + multicenter validation and to establishing ongoing monitoring and recalibration procedures.
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	Emphasized integration with hospital information systems to achieve temporal stability and usability in real-world deployment. Outlined future directions for dynamic/continually updated models and incorporation of intraoperative/postoperative dynamic data. Location in the revised manuscript: (Manuscript) Discussion — Limitations (new “Temporal Validity” paragraph), pages 33, lines 669-678. Reviewer #1 — 5. Lines 901-909 (Claims of novelty and clinical impact are too strong and insufficiently supported)- page33, lines 664-668. Comment: “The statements about the study’s novelty and clinical impact are overly strong without sufficient evidence. Please add citations to early prediction/early warning systems and temper—or provide data to support—claims about assisting physicians.” Response: Thank you for this important comment. We have tempered the language, added background citations, and clarified the scope: 1. In the Discussion, we revised statements about “clinical impact/assisting physicians” to a cautious, hypothesis-generating framing, avoiding definitive claims beyond the presented evidence. 2. We softened and forward-looking-framed remarks on “clinical integration,” emphasizing that prospective evaluation, workflow-integration studies, and external validation are required before routine use. 3. We focused future work on dynamic, continuously updated prediction frameworks and on temporal and multicenter validation to ensure reliability across time and institutions (see the replacement text below). Actions taken: Tone down & clarify boundaries: Deleted/rephrased overly strong language; changed phrases like “assist physicians” to potential/conditional wording. Add citations: Inserted representative references on early prediction/early warning systems and perioperative AKI risk tools in the relevant Discussion paragraph(s) to support context and comparisons. Prospective roadmap & validation: Added forward-looking statements on dynamic models, temporal/multicenter validation, and information-system integration (see replacement text). Location in the revised manuscript: Discussion — Clinical Implications/Future Work, page33, lines 664–668. Reviewer #1 — 6. (Overstated claims about novelty and clinical impact) Comment: The wording on “clinical impact/assisting physicians” should be tempered, and citations to early warning/EMR decision-support systems should be added to provide adequate context and support. Response: Thank you for this important suggestion. We have toned down the language and strengthened the evidence base as follows: Reframed statements about “clinical impact/assisting physicians” as potential contributions and hypothesis-generating, avoiding definitive claims beyond the current evidence; Added targeted evidence linking perioperative preventive measures (e.g., minimizing nephrotoxic exposure, maintaining hemodynamic stability, optimizing volume management) to AKI risk reduction; Incorporated representative studies on early AKI warning systems and EMR-based decision-support to situate our work within the existing literature; Positioned our model as a surgery-specific risk identification module with potential for integration into existing platforms, while emphasizing that prospective evaluation and workflow-integration studies are prerequisites for any routine clinical use. Actions taken: Replaced the original paragraph with a version that uses tempered wording and additional citations; Added conditional, forward-looking phrasing (e.g., “may/has the potential to”) in adjacent sentences, and pointed to the need for external/temporal and multicenter validation as the next steps. Location in the revised manuscript: (Manuscript) Discussion — Clinical Implications/Future Work, page 33, lines 679–688. Reviewer #1 — 7. (Language & formatting: typos, spacing, inconsistent styles)
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	Comment: Please correct typographical/spacing errors and ensure consistent formatting that complies with PLOS ONE style (templates, file naming, figures/tables, references, headings, and supplementary materials). Response: Thank you for pointing this out. We conducted a comprehensive language and formatting sweep and aligned the manuscript with PLOS ONE requirements. Key updates are summarized below.
Additional Information:	
Question	Response
Financial Disclosure Enter a financial disclosure statement that describes the sources of funding for the work included in this submission. Review the submission guidelines for detailed requirements. View published research articles from PLOS ONE for specific examples. This statement is required for submission and will appear in the published article if the submission is accepted. Please make sure it is accurate. Funded studies Enter a statement with the following details: • Initials of the authors who received each award • Grant numbers awarded to each author • The full name of each funder • URL of each funder website • Did the sponsors or funders play any role in the study design, data collection and analysis, decision to publish, or preparation of the manuscript? Did you receive funding for this work?	The author(s) received no specific funding for this work.
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Written consent has been obtained.

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The data underlying the results presented in the study are available

The datasets used in this study, including both imaging data and corresponding clinical information of the study subjects, were collected and curated by the authors. All relevant data, along with the source code for model implementation and analysis, will be made publicly available on the authors' GitHub repository to ensure transparency and reproducibility of the study. The repository is available at : <https://github.com/malingshoo/aki>

<i>from (include the name of the third party and contact information or URL).</i> • This text is appropriate if the data are owned by a third party and authors do not have permission to share the data. * typeset	
Additional data availability information:	

Dear PLOS ONE Editor,

We are excited to submit our manuscript entitled “Few-Shot Prediction of Post-Fusion Acute Kidney Injury Using Multimodal Radiomics-Clinical Feature-Based Prototype Networks” for your consideration for publication in PLOS ONE. The research presented in this manuscript has not been submitted elsewhere for publication, in whole or in part, and all authors listed have given their approval of the enclosed manuscript.

In this study, we propose a novel multimodal few-shot prototype network aimed at optimizing the prediction of acute kidney injury (AKI), a major postoperative complication. By integrating 11 preoperative clinical parameters with imaging features extracted by ResNet-50, we demonstrate the effectiveness of our model in early AKI detection. Our approach significantly outperforms clinical-only and imaging-only models, achieving an area under the curve (AUC) of 0.816. Additionally, we utilize SHapley Additive exPlanations (SHAP) for interpretability, identifying key predictive variables such as age, lactate, and creatinine. This work establishes the first application of few-shot learning with prototype networks in postoperative AKI prediction.

We believe that our findings contribute valuable insights to predictive modeling in the medical field, particularly for postoperative risk assessments, and would be of significant interest to the readership of PLOS ONE. We recommend incorporating the 11 standardized preoperative indicators and imaging data into clinical practice to enhance predictive accuracy.

My deep passion and extensive expertise in this field reinforce my confidence that this manuscript will make a valuable contribution to your journal. We sincerely appreciate your time and consideration in reviewing our work, and we eagerly await the reviewers’ feedback. Should you have any additional questions or require further information, please feel free to contact me at the email provided below.

Sincerely,

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Few-Shot Prediction of Post-Fusion Acute Kidney Injury Using Multimodal Radiomics-Clinical Feature-Based Prototype Networks

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Abstracts

To optimize preoperative decision-making, the prediction of acute kidney injury (AKI)—a major postoperative complication—requires early identification. This study developed a multimodal few-shot prototype network based on real-world clinical data from 194 single-center cases. The model incorporated 11 preoperative parameters encoded by a multilayer perceptron (MLP) to weight and integrate imaging features extracted by ResNet-50, with a fixed loss function implemented to address class imbalance. Results demonstrated that the fused model achieved an area under the curve (AUC) of 0.816, significantly outperforming the clinical-only (AUC = 0.578) and imaging-only (AUC = 0.510) models. Interpretability analysis using SHapley Additive exPlanations (SHAP) revealed the influence of key variables (age, lactate, creatinine). Our study demonstrates that multimodal fusion significantly improves generalization performance in data-limited scenarios, establishing the first application of few-shot learning with prototype networks for postoperative AKI prediction. We recommend incorporating 11 standardized preoperative indicators and imaging data into clinical risk assessment to enhance predictive accuracy and utility. Future directions should focus on expanding cohort sizes and refining radiomic feature extraction.

Keywords: Acute kidney injury (AKI), Few-shot learning, Prototypical networks, Multimodal fusion, SHAP value analysis

1. Introduction

Acute kidney injury (AKI) is a clinically critical condition characterized by an acute decline in renal function with concomitant disturbances in fluid, electrolyte, and acid-base homeostasis. **The Kidney Disease: Improving Global Outcomes (KDIGO) criteria define AKI as: 1. Serum creatinine (SCr) rises by $\geq 26.5 \mu\text{mol/L}$ within 48 hours; or 2. SCr increases to >1.5 times baseline within 7 days; or 3. Urine output remains $<0.5 \text{ mL}/(\text{kg}\cdot\text{h})$ for 6 hours[1].**

Postoperative acute kidney injury (AKI) is prevalent among hospitalized patients, particularly following major surgeries. Its occurrence correlates strongly with increased mortality, prolonged hospitalization, elevated risks of chronic kidney disease progression, and higher healthcare costs [2][3]. In high-risk surgical populations, the incidence of acute kidney injury (AKI) is reported to exceed 20%, and it is notably higher in intensive care unit (ICU) patients [3]. The clinical utility of current diagnostic markers is limited by their substantial time lag, which often fails to detect reversible injury during the early stages of AKI, as highlighted by previous studies [2],[4]. Therefore, early identification and risk stratification are critical for implementing effective renal protection strategies.

As artificial intelligence has advanced, an increasing number of studies have attempted to use deep learning or machine learning techniques for preoperative AKI prediction tasks. Most of these models build risk classifiers using algorithms such as logistic regression, random forests, or multilayer perceptrons, with structured

electronic health record (EHR) data as input. However, the accuracy and generalization ability of these models are often limited by traditional approaches, which tend to rely heavily on tabular features while neglecting unstructured data, such as changes in patient images and physiological variations [5]. This leads to prediction bias and performance imbalances, as traditional models are more likely to predict the majority class during training. Postoperative AKI, a low-incidence but high-risk event, is particularly affected by this issue due to the significant imbalance between positive and negative samples, especially in real-world hospitalized populations. In China, acute kidney injury (AKI) has a high incidence among hospitalized patients, with an estimated prevalence of 20% [6].

Furthermore, most current models lack clinical interpretability. In real-world medical practice, clinicians often require clear explanations of a model's rationale before trusting algorithm-assisted judgments. Conventional "black-box" AI models typically provide insufficient decision transparency, which limits their adoption and deployment in clinical settings[7].

Additionally, unbalanced data remains a major challenge in postoperative AKI risk prediction, as the number of AKI cases often exceeds that of non-AKI patients. This imbalance leads to overfitting and misclassification in traditional machine learning algorithms, which favor the majority class. In clinical settings, such imbalance can significantly undermine model stability and dependability. Furthermore, AKI is a multifactorial condition that involves complex interactions

between factors such as anesthetic procedures, intraoperative blood loss, postoperative infections, and electrolyte imbalances. Therefore, a more comprehensive approach integrating postoperative medical imaging with tabular clinical data is essential for accurate risk identification and etiological modeling.

Few-shot learning (FSL) has increasingly drawn attention in this context. By enabling classifiers to generalize well from only a handful of labeled examples, FSL addresses both the high cost of annotation and the scarcity of medical data. In particular, metric-learning frameworks such as Prototypical Networks (PNs) infer the class distribution of novel samples from a small support set, making them especially well suited to low-incidence, high-risk clinical prediction tasks like postoperative AKI.

In this study, a multimodal small-sample prototype network framework combining preoperative medical imaging and structured form data was built for the problem of postoperative AKI prediction using actual clinical data from **a single center (the Second People's Hospital of Sichuan Province)**. A separate multilayer perceptron (MLP) with weighted fusion encodes the structured form data, while a pre-trained Residual Network (ResNet-50) model extracts features from the image data. We describe the You Only Look Once (YOLO) labeled region cropping mechanism to enhance the efficacy of picture representation by improving the efficacy of image features. Associated loss function was created to enhance the model's recognition capabilities on a limited number of sample classes in order to

address the issue of data imbalance. In addition to increasing the model's accuracy in identifying postoperative AKI risk, this study offers a workable route and theoretical foundation for future high-dimensional medical multimodal modeling and illness prediction on a small number of samples.

In summary, this study has contributed to the predictive modeling of postoperative acute kidney injury (AKI) in a number of noteworthy and creative ways, including:

1. Our study addresses a critical gap in AKI prediction research—the limited diagnostic dimensions—by combining clinical indicators and imaging data with strict adherence to KDIGO diagnostic criteria.

2. Minimally invasive renal surgery—particularly robot-assisted and laparoscopic partial nephrectomy—has gained widespread global adoption in recent years. Compared with open surgery, these approaches achieve comparable oncologic outcomes while offering fewer complications and faster recovery [8], and long-term follow-up shows no significant differences in cancer control or renal function preservation between laparoscopic and open partial nephrectomy [9]. In China, although robotic systems are available in large centers, most hospitals still rely primarily on laparoscopic techniques for both partial and radical nephrectomies. Against this backdrop, laparoscopic renal procedures introduce AKI-relevant stressors—pneumoperitoneum-related hypoperfusion and ischemia-reperfusion during hilar clamping—rendering patients more susceptible to AKI, which occurs in

15–25% of partial nephrectomies by KDIGO criteria [10][11] and is linked to long-term renal decline and new-onset CKD [12][13]. These features highlight a gap in existing, largely generic AKI prediction models—often developed in cardiac or heterogeneous surgical cohorts—which seldom incorporate procedure-specific variables such as pneumoperitoneum pressure and warm ischemia time, underscoring the need for a model tailored to laparoscopic renal surgery [14].

3. Our few-shot prototype network integrates multimodal data (preoperative static variables, early physiological trends, and imaging), significantly improving early-warning performance in limited-data scenarios.

4. Model development used **real-world, single-center data from Sichuan Provincial Second People’s Hospital**, addressing regional gaps in generic AKI models by incorporating local epidemiological, environmental, and healthcare system factors.

5. SHAP interpretability analysis enhances clinical utility by visualizing risk factors, enabling physicians to understand model predictions and improve real-world adoption of this intelligent system.

In conclusion, this study offers a theoretical foundation and workable technical support for the early detection and tailored treatment of postoperative AKI in resource-constrained areas through the development of a practical, interpretable, and geographically flexible postoperative AKI prediction framework.

2. Theoretical Basis and Research Progress

2.1. Advances in Postoperative AKI Prediction Research

Research in clinical renal protection management has focused on the prediction and early detection of postoperative acute kidney injury (AKI), one of the frequent consequences in the perioperative period. The AKI-EPI study provides a comprehensive analysis of the epidemiology of AKI in critically ill patients across multiple centers [15]. Based on imaging histological findings and structured clinical variables, AKI prediction algorithms may currently be generally divided into two groups.

Clinical feature-based prediction techniques typically rely on conventional machine learning models, such as logistic regression or random forests, which utilize basic preoperative data, intraoperative physiological markers, and perioperative laboratory results. For instance, Liu et al. developed a logistic regression model to predict AKI following liver transplantation using preoperative age, blood pressure, and creatinine levels, achieving an AUC of 0.78 on the validation set, demonstrating acceptable clinical feasibility [16]. Rank et al. reported that a random forest model achieved the highest performance in predicting post-cardiac surgery AKI, with an area under the curve (AUC) of 0.88 and a sensitivity of 0.82 [17]. Although these methods are straightforward to implement and highly interpretable, they struggle to capture complex nonlinear relationships within the data and thus often have limited predictive accuracy, particularly when variable interactions are significant or when manually constructing higher-order features is difficult.

The main goals of imaging-histology techniques are to identify patients' possible risk of renal damage using multidimensional vectors and to extract deep quantitative information from medical images (such as postoperative CT and ultrasound). These techniques can improve the ability to measure the risk of AKI in preclinical stages and aid in the clinical detection of “non-visible” early pathological changes on imaging [18]. Zhou et al. developed a radiomics model to predict AKI following heart surgery by extracting renal texture features from postoperative abdominal CT images. The model demonstrated good image-assisted discrimination, achieving an AUC of 0.81 in external validation [19]. However, these techniques are subject to specific technical thresholds in their widespread promotion and use because of their heavy reliance on image quality, segmentation accuracy, and annotation consistency, as well as the absence of consistent standards in multicenter settings.

Current methods for predicting postoperative AKI still encounter several challenges during model development, including small overall sample sizes, class imbalance, and limited multimodal feature-fusion capabilities. Although many studies report a pronounced imbalance between positive and negative cases, the more fundamental issue is the insufficient total number of samples. This scarcity makes training vulnerable to small-sample effects, leading to unstable parameter estimates and reduced generalizability. In particular, even when class ratios are not severely skewed, underrepresentation of minority classes (e.g., non-AKI) can impair model reliability and sensitivity in clinical applications. To address these issues—small

samples, class imbalance, and weak multimodal fusion—we propose a prototypical network approach that integrates postoperative imaging histology with clinical features within a few-shot learning framework, thereby enhancing early detection performance and model robustness for postoperative AKI.

2.2. Multimodal Radiomics Feature Fusion

Recent years have seen a widespread application of multimodal feature-fusion techniques in the field of medical risk prediction, particularly for tasks with high clinical importance like kidney lesions and surgical sequelae. Radiomics is a technique that helps with risk stratification and disease prediction judgment by extracting high-dimensional and quantitative features from medical images (such as CT, MRI, and ultrasound) that can detect early pathological changes that are undetectable by traditional visual assessment. Gillies et al. introduced the concept of radiomics, emphasizing the extraction of quantitative features from medical images for decision support [20]. Information like structural alterations, perfusion status, and kidney interstitial features are frequently seen in postoperative images as slight texture or density changes. Imaging histology may be able to serve as a foundation for identifying these asymptomatic phases, which is important for the prediction of postoperative AKI.

In an effort to provide a more comprehensive risk prediction framework, an increasing number of studies have attempted to simultaneously model radiomics features with structured clinical characteristics (such as serum creatinine, lactate, and

urine output changes). Additionally, some research has investigated fusion architectures based on dual-branch neural networks for imaging and tabular data, which enhance the model's ability to incorporate cross-modal heterogeneous data through adaptive weighting and attention mechanisms [21].

Although current research has preliminarily verified the potential of multimodal fusion strategies for complex prediction tasks such as postoperative AKI, several practical challenges remain: First, dimensional disparities and inconsistent expression mechanisms across modalities often lead to semantic misalignment and information redundancy during fusion. Second, nonstandardized medical imaging acquisition protocols frequently introduce image quality variations and segmentation errors that compromise feature stability. Finally, the complex architectures of multimodal models typically demand large training datasets and high annotation precision, limiting their generalizability in small-sample scenarios. To address these limitations, future research should focus on developing efficient, generalizable, lightweight fusion strategies to enhance model stability and cross-scenario adaptability.

2.3. Few-Shot Learning Methods

In recent years, few-shot learning (FSL) has gained popularity in medical image analysis. Its goal is to enable effective modeling with very few labeled samples. Few-shot learning is particularly valuable when medical image data acquisition is challenging and manual annotation is costly. The primary objective is to achieve rapid generalization and accurate discrimination of new sample categories by constructing

metric spaces or refining meta-learning strategies using limited support samples [22].

Prototypical Networks (PN), a widely used metric-learning approach in FSL, classify samples by calculating feature means of support samples as class prototypes. Query samples are then compared to these prototypes using Euclidean distance. This architecture has demonstrated significant success in various medical applications including breast cancer screening [23], brain tumor classification [24], and fundus lesion detection [25], owing to its simplicity, training stability, and strong generalization capability under data scarcity.

Furthermore, Matching Networks [26] enhance small-sample structural discrimination by considering global matching relationships between query and support samples, while Relation Networks [27] improve fine-grained lesion recognition through learned nonlinear similarity functions. Recent studies have also integrated FSL with multimodal fusion techniques, such as jointly embedding structured clinical data and imaging features into a shared representation space [28], to boost prediction accuracy via collaborative modeling. These fusion strategies have progressively expanded FSL clinical applications in areas like diagnostic assistance and postoperative risk prediction.

3. Methodology

3.1. Study Subjects and Data Collection

3.1.1. Participant Eligibility Criteria

This retrospective study analyzed patients with renal space-occupying lesions

who underwent laparoscopic renal surgery at **Sichuan Provincial Second People's Hospital**. Archived clinical and imaging data were accessed on 1 May 2025. All participants were selected according to strict inclusion and exclusion criteria to ensure population homogeneity and minimize confounding factors. No identifiable data were accessed during or after data collection, and the authors had no access to any information that could identify individual participants. All data were anonymized to ensure confidentiality.

Inclusion Criteria:

- (1) Patients with radiologically confirmed renal space-occupying lesions (via ultrasound, urologic CT, or intravenous urography (IVU)).
- (2) Patients who underwent either laparoscopic partial nephrectomy or radical nephrectomy at our institution, with all surgeries performed by the same experienced chief urologist.
- (3) Adult patients aged > 18 years.
- (4) Patients who had serum creatinine levels measured at least once preoperatively and within 7 days postoperatively.
- (5) Patients with complete baseline clinical data.

Exclusion Criteria:

- (1) Patients with end-stage renal disease (ESRD).
- (2) Patients undergoing concomitant surgeries.
- (3) Patients with a history of renal transplantation.

(4) Patients with other major organ diseases, malignancies, or undergoing chemotherapy.

(5) Patients missing critical data.

This study might more accurately determine the incidence of acute renal injury following surgery while accounting for the impact of significant confounding variables by screening the study participants using the aforementioned criteria. The procedure for selecting study participants based on predetermined inclusion criteria is shown in **Fig 1**.

3.1.2. Multimodal Data Acquisition

Numerous machine learning and deep learning models using preoperative clinical features have been developed to predict postoperative acute kidney injury (AKI), but these unimodal approaches still have limitations. First, most studies rely solely on clinical variables—such as biochemical indices, vital signs, and medical history—to construct risk scores or train models. For example, Yoon et al.’s CMC-AKIX model, built on 38 preoperative clinical variables, achieved an AUC of 0.83 in a retrospective cohort; however, its applicability is largely confined to specific surgical types and requires larger-scale external validation before it can be generalized to wider clinical settings **Error! Reference source not found..** Furthermore, a systematic evaluation of 150 AKI prediction models highlighted substantial variability in study design, cohort composition, AKI definitions, and predictor variables, leading to high risk of bias and heterogeneity, which limit the models’ clinical utility [30]. More importantly,

clinical data alone may fail to capture microanatomical or functional changes apparent in preoperative imaging, which often manifest before biochemical markers such as serum creatinine become abnormal.

To address these limitations, this study integrates two complementary data types—preoperative clinical indicators and medical imaging—through a multimodal learning framework. Multimodal learning, a machine learning paradigm that combines heterogeneous inputs from diverse sources (e.g., images, text, tabular data), enhances model performance by exploiting intermodal complementarity [31]. Specifically, we propose and quantify an imaging-derived anatomical parameter (“**solitary kidney status**”) and integrate it with clinical indicators in our predictive model.

The imaging-derived feature “solitary kidney status” holds significant clinical importance. A retrospective study demonstrated that patients with congenital or surgically acquired solitary kidneys faced substantially higher AKI risks during procedures with potential renal injury, such as percutaneous angiography. In a cohort of 3,456 angiography patients, Mayo et al. reported an odds ratio (OR) of 1.9 (95% CI 1.3-2.8) for AKI occurrence in the solitary kidney group versus the two-kidney group [32]. Additionally, Wang et al.'s cohort study of 786 renal tumor patients revealed a postoperative AKI incidence of 24.7% in patients with preoperative unilateral nephrectomy history, significantly higher than the 12.5% incidence in patients with two kidneys ($P < 0.01$) [33]. These findings confirm that solitary kidney status

effectively reflects interindividual variability in renal functional reserve. Incorporating this parameter into postoperative AKI prediction models could enhance high-risk patient identification accuracy and guide personalized preoperative interventions.

This study developed a multimodal fusion strategy for postoperative AKI risk prediction by integrating feature representations from two complementary data modalities: medical imaging and clinical indicators. For the clinical modality, we incorporated significant preoperative clinical parameters (blood glucose, creatinine, lactate, and urea nitrogen) based on international guidelines, along with the imaging-derived parameter “**solitary kidney status.**” For the imaging modality, we employed a pre-trained ResNet-50 network to automatically extract deep spatial features from preoperative images, including the critical “solitary kidney status” assessment. To comprehensively capture patients' anatomical and physiological status, we performed cross-modal fusion of these features using both feature concatenation and a gated weighting module, generating a unified multimodal representation. Our data integration approach not only addresses limitations of unimodal analysis but also establishes a robust foundation for postoperative AKI prediction.

3.1.3. Clinical Feature Selection and Definition

This study utilized several preoperative clinical indicators to develop a predictive model for postoperative acute kidney injury (AKI) in patients undergoing laparoscopic renal surgery. These indicators provide a comprehensive assessment of patients' hemodynamic status, metabolic abnormalities, and renal function, offering

critical information for postoperative AKI risk prediction.

To provide a more comprehensive assessment of patients' baseline renal function and hemodynamic status, the model initially incorporated traditional preoperative markers, including **age, serum creatinine, urea nitrogen, and hemoglobin**. Older patients show increased vulnerability to acute kidney injury (AKI) due to age-related renal function decline. Elevated serum creatinine and urea nitrogen levels typically indicate impaired renal function and consequently higher AKI risk. Hemoglobin levels reflect the patient's anemia status, which may lead to postoperative renal ischemia and consequently increase AKI incidence.

To further enhance the model's sensitivity to preoperative metabolic disorders and tissue perfusion changes, this study innovatively introduced two additional indicators: **preoperative glucose and lactate levels**. Preoperative glucose levels have demonstrated a close association with AKI risk, showing approximately 15% increased risk for every 1 mmol/L elevation in noncardiac surgical patients [34]. Similarly, preoperative lactate levels are strongly associated with severe AKI occurrence, as studies have shown that postoperative lactate levels > 3.3 mmol/L at 12 hours significantly increase severe AKI risk [35].

Two renal function-associated risk factors—**hypertension and diabetes mellitus**—were also included in the analysis. Diabetes mellitus induces renal damage through altered glucose metabolism and is associated with higher postoperative AKI incidence, especially in patients with inadequate glycemic control. Hypertension,

defined as systolic blood pressure ≥ 140 mmHg or diastolic blood pressure ≥ 90 mmHg, is an established risk factor for chronic kidney disease.

Furthermore, **blood urea nitrogen (BUN)** and **erythrocyte-specific volume** were selected as traditional indicators of renal function and hemodynamics for this model. Low erythrocyte-specific volume or perioperative anemia significantly elevates postoperative AKI incidence by reducing tissue oxygen supply, particularly during major surgeries with substantial blood loss, rendering anemic patients more vulnerable to AKI. BUN serves as a crucial glomerular filtration rate marker and, together with serum creatinine, represents a standard monitoring parameter for AKI diagnosis and management [36].

3.2. Data Preprocessing and Augmentation

To preserve sample size and minimize the influence of outliers, missing preoperative clinical values were replaced with the median of each variable. Prior studies on medical datasets have demonstrated that median imputation yields predictive performance comparable to more complex multiple-imputation techniques [37]. To remove unit disparities and facilitate model convergence, Z-score normalization was subsequently applied to all continuous variables [38]. The complete imputation and normalization workflow is provided in **Supporting information (S1 Appendix. Data imputation and standardization)**.

In the imaging pipeline, regions of interest (ROIs) were cropped according to YOLO annotations and rescaled to a uniform resolution. Lightweight geometric and

luminance augmentations were applied to increase data diversity before feature extraction by a pre-trained network (**S2 Appendix. Data Augmentation, Class Balancing, and Feature Extraction**).

Given the pronounced class imbalance in the dataset (132 AKI vs. 62 non-AKI cases), a multimodal framework combining feature-space interpolation, weighted sampling, and class-weighted loss was employed to improve model sensitivity to the minority class.

Furthermore, an anchor-based pseudo-sample generation strategy was proposed to further expand the negative-class feature space while maintaining multimodal semantic consistency (**Fig 2**). This approach utilizes pre-trained feature vectors to guide pseudo-sample generation, thereby enhancing data diversity and model interpretability. The parameter configurations and algorithmic details are described in **Supporting information (S2 Appendix. Data Augmentation, Class Balancing, and Feature Extraction)**.

3.3. Multimodal Feature Fusion Network

This study presents a multimodal feature-fusion framework to effectively leverage complementary information from imaging and tabular data in small-sample clinical contexts, addressing challenges such as data scarcity, modality imbalance, and cross-modal alignment. As shown in **Fig 3**, visual features are extracted from the image branch using a pre-trained **ResNet-50**, yielding 2048-dimensional representations, while structured features are obtained from the tabular branch via a

multilayer perceptron encoder, resulting in 128-dimensional representations. These features are combined into a joint representation. Next, parallel sigmoid gating assigns dynamic weights to each modality, amplifying reliable signals and suppressing noise. The weighted high-dimensional vectors are compressed into 128-dimensional embeddings through a two-layer fully connected network with **batch normalization**, **LeakyReLU** activation, and **dropout**. Finally, **L2 normalization** ensures compatibility with metric learning. In imbalanced postoperative AKI prediction tasks, this lightweight architecture improves sensitivity and generalization while avoiding complex attention mechanisms and extensive hyperparameter tuning.

3.3.1. Feature Extraction Module

To integrate multimodal data, we use a dual-branch feature-extraction architecture.

Image feature encoder: A pre-trained ResNet-50 with YOLO-based ROI cropping provides high-level visual features while minimizing background interference; the design supports flexible fine-tuning for small-sample settings.

Tabular feature encoder: A shallow MLP maps structured clinical variables into a compact representation with ReLU activations and dropout for better generalization. Implementation details are provided in **Supporting information (S3 Appendix. Detailed Design of the Feature Extraction Module)**.

3.3.2. Dynamic Gated Fusion and Dimensionality Reduction

To fuse modalities, we adopt **sample-wise sigmoid gating** that independently

weights each modality, avoiding the sum-to-one competition of softmax-based attention. Informative modalities can thus be jointly amplified, while unreliable inputs are automatically down-weighted. The gated representation is then passed through a **lightweight two-layer projection** to obtain a **compact 128-dimensional embedding**, followed by **L2 normalization** to map embeddings onto the unit sphere and harmonize Euclidean and cosine metrics. In Prototypical Networks, class prototypes are obtained by calculating the simple mean of normalized embedded vectors from the support set, and the normalized prototypes also lie on the unit sphere, ensuring consistency and interpretability in distance metrics between classes [39]. This design requires no extra hyperparameter tuning and improves sensitivity and generalization under small-sample and class-imbalanced conditions (Detailed equations and network parameters are provided in **Supporting information (S4 Appendix. Dynamic Gated Fusion Mechanism and S5 Appendix. Multimodal Feature Fusion and Dimensionality Reduction)**).

3.4. Few-Shot Learning Strategy

3.4.1. Multi-Prototype Construction

In classical **Prototypical Networks (PN)** [39], each class is represented by a single prototype computed as the mean of support embeddings, which may be insufficient to capture **intra-class multimodality** and subcategory structure in real-world data [40]. To address this, we adopt a **KMeans-based multi-prototype** scheme: support-set embeddings of the same class are aggregated into multiple

prototypes that jointly describe the class distribution (**Fig 4**).

The clustering step is used **offline** as an unsupervised aid and does **not** interfere with the supervised training objective; it enriches class structure and improves the stability and accuracy of metric-based inference. For categories with extremely few samples, we apply a **mean-based compensation** to maintain a **fixed number** of prototypes per class, ensuring consistent interfaces during inference. Prior studies also report benefits of multi-prototype representations in settings with pronounced intra-class variation—for example, Zhu et al.’s **Multi-Prototype Prototypical Network** achieves strong results in fine-grained recognition—supporting our design choice [41]. Full construction details are provided in **Supporting information (S6 Appendix. Few-Shot Learning Strategy)**.

3.4.2. Prototype Normalization and Minimum-Distance Selection

To eliminate scale differences and ensure a consistent metric space, **L2 normalization** is applied to all prototypes after construction. During classification, a query’s similarity to a class is defined as the **minimum Euclidean distance** to that class’s prototypes, aligning the query with its **nearest local pattern** within the class and thereby improving recognition accuracy and robustness (**Supporting information: Details are available in S6 Appendix. Few-Shot Learning Strategy**).

Compared to a single-prototype strategy, multi-prototype approaches can more finely capture intra-class multimodal characteristics, with particularly pronounced benefits in medical few-shot scenarios. This aligns with Tang et al.’s Adaptive

462 Prototype Learning (APL), which demonstrated higher accuracy and generalization in
 463 remote-sensing image classification through a multi-prototype design [42]. **Table 1**
 464 summarizes the core differences and advantages between traditional single-prototype
 465 methods and our proposed multi-prototype approach.

Method	Number of Prototypes per Class	Intra-class Feature Representation Ability	Computational Complexity	Handling Method for Insufficient Samples
Prototypical Networks[40]	1	Low	Low	None
Multi-Prototype[42]	≥ 2	Medium	Medium	Simple Mean Compensation
Proposed Method	3	High	Medium	KMeans+Mean Compensation
Matching Networks[43]	N/A	Medium	High	Episodic Training Framework, no fine-tuning required
Relation Networks[44]	N/A	High	High	Learn deep nonlinear metrics, no explicit prototypes required
Transductive Propagation Network[45]	Not Applicable	High	High	Label propagation using the entire query set

466 **Table 1. Comparative Analysis of the Proposed Multi-Prototype Method and**
 467 **Conventional Approaches**

- 468 • Matching Networks employ an attention mechanism to perform pointwise
 469 matching between support-set examples and query samples. Through episodic
 470 training, they enable rapid generalization in few-shot scenarios without

introducing explicit prototypes [26].

- Relation Networks learn a transferable deep metric function (Relation Module) via a two-branch network, outperforming fixed-distance metrics in feature representation and eliminating the need for prototype construction [43].
- Transductive Propagation Network (TPN) applies a graph-propagation strategy during testing to jointly infer labels across support and query sets, effectively mitigating information scarcity in few-shot settings [44].

In contrast, our method retains the efficiency of prototypical networks while compensating for the limitations of single-center representations through multiple prototypes, achieving a balance between performance and computational cost.

3.5. Loss Function Design

To address three key challenges in clinical few-shot multimodal classification—insufficient intra-class diversity, semantic drift of pseudo-samples, and uneven embedding-space distribution—we propose a **multi-objective loss fusion framework**.

First, a multi-prototype classification loss captures the multimodal and multi-peak distributions within each class, effectively representing intra-class heterogeneity beyond traditional single-center prototype networks [41][46]. Second, an anchor alignment loss ensures semantic consistency between generated pseudo-samples and their true category anchors (e.g., SwAV-derived vectors), thereby mitigating semantic drift and enhancing pseudo-sample reliability [47][48]. Third, a batch-hard triplet loss

focuses on the most discriminative positive–negative sample pairs, tightening category boundaries and accelerating convergence while improving inter-class separability [49][50].

Finally, these sub-losses are weighted and combined into a unified training objective following multi-objective optimization principles [51], which balances classification accuracy, semantic alignment, and embedding-space discriminability. Detailed mathematical formulations, parameter configurations, and ablation analyses of each loss component are provided in **Supporting information (S7 Appendix. Loss Function Design)**.

3.6. Training Procedure

The overall training workflow (**Fig 5**) is as follows: first, raw data are standardized and negative-class samples are augmented; then, support and query sets are generated using few-shot episodic sampling. The multimodal feature-extraction module separately processes imaging data with ResNet-50 and tabular data with an MLP, and gated fusion produces 128-dimensional embeddings. K-means clustering generates multiple prototypes per class, and three loss terms are computed and jointly optimized. Model parameters are iteratively updated using the Adam optimizer, and the final multimodal fusion model weights—capable of handling both image and clinical inputs—are saved. The workflow incorporates dynamic class weighting and clinical-range validation to ensure reliability and clinical plausibility in data-scarce scenarios.

4. Model Evaluation

To evaluate our proposed multimodal model, we employed a nested cross-validation strategy with five-fold stratified partitioning to ensure balanced class distributions in each fold. **Fig 6** illustrates the workflow of our evaluation process. This approach was chosen to mitigate bias arising from data imbalance and to ensure robust performance estimates. In each fold, we utilized pretrained submodels—including a multilayer perceptron (MLP) for tabular data, a convolutional neural network (CNN) for imaging data, and our prototype network—to extract features from both modalities. The predictions from these models were then averaged via an ensemble method. Binary classification decisions were made using a global threshold determined by the Youden index to further optimize performance. Model efficacy was quantified using multiple metrics—accuracy, balanced accuracy, precision, recall, F1 score, ROC AUC, and Matthews correlation coefficient (MCC)—computed for each fold and summarized by their mean values.

5. Results

Through systematic multimodal data integration, the fusion-based postoperative AKI prediction model proposed in this study demonstrated significantly greater performance improvements compared to single-modality approaches. The imaging-only model (ResNet-50) achieved an AUC of 0.510 in independent validation, while the clinical data model (incorporating preoperative creatinine, lactate, age, etc.) reached an AUC of 0.578. Although both models exhibited some predictive

capability, they showed clear limitations. By innovatively combining these two modalities with prototype learning, the final model achieved substantially enhanced performance (AUC = 0.816).

Through SHAP-based interpretability analysis, this study effectively reveals the clinical feature contributions underlying the model's decision-making process, enabling clinicians to intuitively understand the biological basis of predictions and increasing their trust in the model. This transparent interpretation mechanism allows physicians to both comprehend general decision patterns and validate prediction rationality for individual cases. Consequently, it significantly enhances the efficiency and reliability of preoperative risk assessment in human-AI collaboration, facilitating clinical implementation of the AI model.

Although the representativeness of our sample and the generalizability of derived thresholds require further improvement, our multimodal fusion framework has demonstrated significant clinical value. In future work, multicenter, large-scale cohorts encompassing diverse surgical procedures will expand sample coverage and enable more granular subgroup analyses. Additionally, implementing dynamic thresholding strategies tailored to individual patient characteristics and clinical contexts will further enhance personalized adaptability. Concurrent integration of intraoperative real-time physiological monitoring data and richer imaging and histological features is expected to establish a closed-loop system uniting preoperative risk assessment with intraoperative early warning. These advances will accelerate the

translation of our model from research validation to routine clinical application, providing intelligent support for precise perioperative decision-making.

5.1. Model Performance

Using five-fold cross-validation, the postoperative acute kidney injury (AKI) prediction model developed in this study demonstrated robust predictive performance. The model exhibited stable prediction across different data splits, with an average AUC of 0.816 ± 0.036 . The overall ROC curve (**Fig 7**) remained consistently above the random classification baseline, while individual fold AUC values ranged from 0.764 to 0.863, with the fourth fold achieving optimal performance (AUC = 0.863, accuracy = 0.811). The model showed excellent sensitivity (recall 0.788 ± 0.074), enabling effective identification of AKI-positive cases and reducing underdiagnosis risk in clinical practice. However, precision (0.729 ± 0.038) exhibited greater variability, particularly in fold 1 where it decreased to 0.679, indicating the need for further optimization to reduce false positives in certain data subsets. Detailed performance metrics are shown in the per-fold confusion matrices (**Fig 8**) and ROC curves (**Fig 9**).

The optimal threshold determined by Youden's index was 0.332, which demonstrated good performance in balancing sensitivity and specificity. However, its applicability requires careful consideration in specific populations or clinical contexts. Future studies should explore threshold adjustments tailored to particular medical scenarios (e.g., high-risk surgical patients or elderly populations) to further optimize predictive performance across different applications. From a clinical decision-making

perspective, comprehensive analysis of ROC curves and confusion matrices supports the model's potential value for preoperative risk stratification and postoperative intervention planning. Additional research should focus on external validation and real-world clinical evaluation to enhance the model's generalizability and practical utility.

5.2. SHAP-based Feature Importance

In this study, the SHAP (SHapley Additive exPlanations) method was applied to evaluate model interpretability and analyze the contribution of each preoperative clinical variable to AKI prediction (**Fig 10**). SHAP values quantify the average influence of individual features on model outputs, revealing potential biological mechanisms behind the model's decisions. Ranked by global mean absolute SHAP values:

- **Preoperative serum creatinine:** the most predictive indicator, reflecting the baseline renal function's decisive role in postoperative AKI risk and underscoring the key clinical utility of traditional renal markers in early warning.
- **Age:** increased susceptibility to AKI in elderly patients highlights the need to consider physiological aging in preoperative assessments.
- **Preoperative lactate:** signifies the important influence of perioperative tissue hypoperfusion and metabolic disturbances on AKI pathogenesis.

Moreover, routine preoperative biochemical markers such as AST and blood urea

nitrogen contributed moderately to prediction, whereas chronic comorbidities—hypertension, diabetes mellitus, and solitary kidney status—showed lower SHAP values, likely reflecting the small number of solitary kidney cases in our cohort and the model’s limited learning of this feature’s potential impact.

Overall, SHAP analysis not only confirmed the model’s effective capture of key clinical predictors but also provided clear targets for clinical intervention: preoperative monitoring and optimization of creatinine, lactate, and age-related risk factors are recommended to support precise risk stratification and the development of effective preventive strategies to reduce postoperative AKI incidence.

6. Discussion

This study's postoperative AKI prediction algorithm offers significant methodological innovation. Traditional deep learning techniques perform poorly on small-sample data because of the short data samples, particularly when dealing with unbalanced data that is prone to overfitting. Because of this, despite having little data, Few-Shot Learning (FSL), a crucial area of study in artificial intelligence, may learn efficiently and produce precise predictions. This study successfully avoids the overfitting issue in the conventional methods and effectively enhances the model's generalization capacity by incorporating small-sample learning and data enhancement strategies, which guarantees the model's strong performance under restricted samples. The model maintains clinical value while achieving excellent predicted accuracy through the integration of imaging data with routine preoperative clinical markers.

The model's operability in the clinical setting is further enhanced by the fact that, according to systematic analysis, it only has to include 11 standard preoperative examination indicators to attain good predictive performance. In the meantime, the use of SHAP interpretative analysis guarantees the interpretability of model outcomes and successfully improves the transparency of model decision-making.

Our study creatively added the clinical feature of “solitary kidney status” as an independent component in the selection of characteristics, in addition to combining the imaging and tabular data. We believe that the potential contribution of this feature to the model should not be overlooked, even though the SHAP analysis revealed a small effect of this factor on the model. This may be related to the small sample size of patients with solitary kidneys (only 7 out of 194 cases), and that this feature is based not only on clinical data (such as the patient's history) but also may be associated with imaging findings (such as the kidney's structural features), providing a multidimensional perspective that is especially relevant in certain high-risk patient populations. The solitary kidney factor can become increasingly significant in the model as the sample size grows. Additionally, imaging data were crucial to our investigation. The current preoperative indicators can already be used for effective prediction, even though detailed imaging indicators like tumor size and location were left out. Future optimization of the imaging feature selection process can also be carried out, and additional structural and functional imaging data can be integrated to enhance the model's accuracy and clinical applicability.

Furthermore, the tabular data selection process adhered to the current international standards, which include diabetes [52] and hypertension [53] as predictors. According to earlier research, the two main chronic conditions influencing renal function are diabetes mellitus and hypertension, both of which have long-term impacts on the kidney through chronic pathways [54][55]. Consequently, a more thorough clinical viewpoint for forecasting postoperative AKI is offered by the model's incorporation of preoperative blood pressure and glucose levels, as well as the presence of diabetes and hypertension.

From a pathophysiological point of view, the model's primary predictors not only support current clinical knowledge but also offer new insights into the possible mechanisms through which preoperative metabolic status (such as lactate levels) influences the development of postoperative AKI. With regard to clinical translation, the current model differs from earlier prediction systems that depend on specialized testing methods; it uses standard preoperative screening indicators, which increases its generalizability. Specifically, the model's probabilistic risk score values can be easily incorporated into current clinical decision pathways, significantly aiding physicians in making preoperative decisions.

Nonetheless, there are still certain restrictions on the current study. In addition, the current model only provides static preoperative prediction and excludes intraoperative and postoperative dynamic monitoring data, which somewhat affects the real-time and accuracy of the prediction. The single-center data source limits the

model's generalization ability, particularly the validity in special populations (such as children, patients with a single kidney, or patients with severe renal impairment), which has not yet been confirmed. Future research should focus on developing dynamic and continuously updated prediction models to enhance robustness and generalizability across diverse surgical populations. In addition, establishing **temporal and multicenter validation frameworks** will be crucial to ensure model reliability over time and across institutions.

Additionally, the **temporal validity** of the model should be acknowledged. As the model was developed from historical single-center data, future changes in perioperative workflows, surgical techniques and instruments, anesthesia and hemodynamic management, and laboratory testing methods may alter the distribution of input features and the incidence of AKI, potentially leading to **temporal performance drift**. To mitigate this risk, we plan to perform **temporal external and multicenter validation**, and to establish **regular performance monitoring and recalibration procedures**. Furthermore, integration with hospital information systems will be advanced to ensure the model's temporal stability and clinical applicability in real-world practice.

From a clinical perspective, the **potential usefulness** of our model lies in its contribution to **perioperative preventive medicine**. By accurately identifying high-risk patients before surgery, clinicians may implement **targeted preventive strategies**, such as minimizing nephrotoxic drug exposure, maintaining hemodynamic

stability, and optimizing volume management—to **potentially reduce postoperative AKI incidence** [3]. Furthermore, building upon advances in early AKI warning systems and EMR-based decision-support tools [7][56], integration of our **surgery-specific model** into these platforms could facilitate **real-time risk monitoring and proactive intervention**, providing a conceptual and practical foundation for intelligent postoperative complication management.

7. Acknowledgments

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850 **Figure information**

851 **Fig 1. Participant Eligibility Criteria.** Participant screening and grouping process.

852 Researchers screened individuals according to predefined inclusion criteria, yielding a
853 final study cohort of 194 subjects.

854 **Fig 2. Three-step data augmentation strategy for negative-class samples.** first,
855 diverse negative pseudo-samples are generated using Anchor vectors; second,
856 Euclidean distances among negative samples are computed to identify nearest
857 neighbors for interpolation, enhancing local diversity; finally, the pseudo-samples are
858 combined with real negative samples.

859 **Fig 3. Multimodal Feature Fusion Network.**

860 **Fig 4. Process for Generating Multiple Prototypes.** Samples are first grouped by
861 class, then K-means clustering is used to compute centroids for each class; if a class
862 has insufficient samples, mean imputation is applied; finally, the centroids are
863 L2-normalized to produce the set of multiple prototypes.

864 **Fig 5. Training Workflow.**

865 **Fig 6. Model Evaluation Workflow.**

866 **Fig 7. Overall ROC Curve**

867 **Fig 8. Five-fold Cross-validation – Confusion Matrix**

868 **Fig 9. Five-fold Cross-validation – ROC Curves**

869 **Fig 10. Feature Importance for Postoperative AKI Prediction**

Table 1. Comparative Analysis of the Proposed Multi-Prototype Method and Conventional Approaches

Supporting information

S1 Appendix. Data imputation and standardization

Missing preoperative clinical values were replaced with column-wise medians to preserve sample size and reduce the influence of outliers. Comparative studies have shown that median imputation performs comparably to more complex multiple imputation methods. To further mitigate potential overfitting from fixed imputed values, the prototype network adopted episodic training, dynamically reorganizing support sets to expose the model to varied sample combinations.

After imputation, predefined clinical variables were standardized using scikit-learn's StandardScaler, which computed the mean and standard deviation on the training set and transformed all variables to a standard normal distribution (zero mean and unit variance). This normalization eliminated scale disparities, stabilized training, and improved model convergence and generalization.

To ensure reproducibility, the normalization parameters were serialized and reused during inference, completing the preprocessing pipeline from raw clinical data to model-ready inputs.

S2 Appendix. Data Augmentation, Class Balancing, and Feature Extraction

To address the pronounced imbalance between AKI-positive ($n = 132$) and negative ($n = 62$) cases, two complementary augmentation strategies were designed

for image and tabular modalities within the multimodal framework.

In the image branch, regions of interest (ROIs) were cropped from YOLO-based annotations and rescaled to original pixel dimensions. Lightweight geometric and luminance transformations—including small random rotations ($\pm 5^\circ$), horizontal flips, and brightness/contrast adjustments—were applied online before resizing all images to 224×224 pixels. A pre-trained ResNet-50 (with its classification head removed) was then used to extract 2048-dimensional deep features for downstream fusion. These procedures expanded data diversity and enhanced model generalization under limited-sample conditions.

In the tabular branch, a SMOTE-inspired interpolation strategy was adopted to enrich the minority class. Positive samples were randomly paired and linearly interpolated within a normalized feature space to synthesize realistic minority examples, while negative samples were interpolated along the prototype network's decision boundary to generate harder negatives. During training, a WeightedRandomSampler and class-weighted loss function were jointly employed to balance class distributions and improve sensitivity to AKI cases.

Furthermore, to improve semantic consistency across modalities and overcome the instability of traditional feature-space generation methods (e.g., GAN or Mixup), we introduced an anchor-based pseudo-sample generation strategy (Fig 2). This three-step approach involved:

- Generating diversified pseudo-samples using AnchorFeatureGenerator initialized

- 912 from SwAV-pretrained anchor vectors and random noise;
- 913 ● Calculating Euclidean distances among standardized negative-class features to
- 914 derive an adaptive interpolation threshold (25th percentile);
- 915 ● Mixing pseudo-samples with real negative samples for multi-prototype
- 916 construction and loss computation.

917 This strategy effectively expanded the coverage of the negative-class feature

918 space, enhanced sample diversity, and improved model interpretability.

919 **S3 Appendix. Detailed Design of the Feature Extraction Module**

- 920 ● **S3.1 Image Feature Encoder:** A pretrained ResNet-50 model is used as the
- 921 visual backbone. The final classification layer is removed and global average
- 922 pooling (GAP) is retained to produce 2048-dimensional feature vectors.

923 For ROI cropping based on YOLO annotations, normalized bounding-box

924 coordinates are converted to pixel coordinates; the cropped regions are then

925 uniformly resized to 224×224.

- 926 ● **S3.2 Tabular Feature Encoder:** The encoder adopts a two-layer multilayer
- 927 perceptron (MLP), with each layer consisting of a linear transformation, ReLU
- 928 activation, and Dropout (dropout rate = 0.2).

929 The inputs are Z-score-standardized clinical features. The first layer maps the

930 original dimensionality to 128 dimensions, and the second layer performs a

931 nonlinear transformation within the same dimensional space.

932 This design aims to enhance feature expressiveness while mitigating overfitting

933 in small-sample scenarios.

934 ● **S3.3 Module Parameters and Compatibility**

935 Output dimensionality of the image encoder: 2048

936 Output dimensionality of the tabular encoder: 128

937 The two modalities exhibit good compatibility in terms of dimensionality and
938 distribution, providing a unified input for the subsequent fusion module.

939 **S4 Appendix. Dynamic Gated Fusion Mechanism**

940 To achieve effective integration of visual and structured data under
941 limited-sample constraints, we developed a dynamic gated fusion mechanism that
942 adaptively assigns modality-specific contribution weights according to each sample's
943 feature distribution. This approach maintains robust cross-modal interactions while
944 mitigating information imbalance and overfitting risks through sample-adaptive
945 weight allocation. The complete module architecture operates as follows:

946 **Feature splicing:** the 2048-dimensional visual features(f_{img}) output from the image
947 encoder are spliced with the 128-dimensional structured features output from the table
948 encoder(f_{tab}) in the feature dimension to obtain a (2048+128)-dimensional joint vector:

949
$$f_{concat} = [f_{img} ; f_{tab}] \in \mathbb{R}^{2176} \quad (1)$$

950 ● **Two-layer fully-connected:** the spliced vectors are sequentially passed through a
951 two-layer fully-connected network, each layer coupled with ReLU activation, to
952 capture higher-order interactions across modalities:

953
$$h_1 = \text{ReLU}(W_1 \cdot f_{concat} + b_1), \quad h_2 = \text{ReLU}(W_2 \cdot h_1 + b_2) \quad (2)$$

954 In the formula: $W_1 \in \mathbb{R}^{128 \times 2176}, W_2 \in \mathbb{R}^{2 \times 128}$

955 ● **Sigmoid output dual-channel weighting:** replacing the traditional softmax
956 attention mechanism with independent sigmoid gating, independently weighted
957 multimodal feature fusion, applying sigmoid directly to the two output channels:

$$958 \quad [g_{\text{img}}, g_{\text{tab}}] = \sigma(W_3 \cdot h_2 + b_3), g_{\text{img}}, g_{\text{tab}} \in (0, 1) \quad (3)$$

959 Where g_{img} is used to weight visual features and g_{tab} is used to weight
960 structured features.

961 S5 Appendix. Multimodal Feature Fusion and Dimensionality Reduction

962 After obtaining the adaptive gating weights for the image and tabular branches,
963 the next step is to reconstruct and fuse the two feature sets to generate a
964 low-dimensional embedding rich in cross-modal information.

965 ● **Weighted feature reconstruction:** features are dynamically adjusted so that
966 important features are more prominent and unimportant features are suppressed.
967 Here for a single sample, let its visual feature vector be $f_{\text{img}} \in \mathbb{R}^{2048}$ and
968 structured feature vector be $f_{\text{tab}} \in \mathbb{R}^{128}$, the corresponding gating outputs are
969 scalar g_{img} and g_{tab} , which are first applied separately:

$$973 \quad \tilde{f}_{\text{img}} = g_{\text{img}} \times f_{\text{img}}, \tilde{f}_{\text{tab}} = g_{\text{tab}} \times f_{\text{tab}} \quad (4)$$

970 This allows for adaptive amplification or suppression of a certain way of
971 information according to the sample, preventing noise modes from affecting the
972 final representation.

974 ● **Fused Feature Normalization:** To avoid dominating the model learning when

the value range of a feature is large after fusion, we splice the weighted two vectors by dimension to get $f_{\text{used}} \in \mathbb{R}^{2176}$, and then apply BatchNorm to normalize the spliced vectors:

$$f_{\text{norm}} = \text{Batch Norm}(f_{\text{used}}) \quad (5)$$

to eliminate differences in the numerical scales of the two-way features and to mitigate cross-modal internal covariate bias.

- **Dimensionality Reduction Projection (DRP):** normalized features are first activated by LeakyReLU, then Dropout regularization is done, and finally mapped to the low-dimensional space:

$$h = \text{LeakyReLU}(f_{\text{norm}}), h' = \text{Dropout}(h, p=0.2), z = W_{\text{proj}} \cdot h' + b_{\text{proj}} \quad (6)$$

where $z \in \mathbb{R}^{128}$ is the final multimodal embedding. the negative slope of LeakyReLU ensures the preservation of negative features, and Dropout provides additional regularization in small sample scenarios.

S6 Appendix. Few-Shot Learning Strategy

S6.1: Multi-Prototype Construction Details

In order to create several prototype centers (Multi-Prototypes) for each category, we first extract the feature representations of each category from the support set using the KMeans approach. The KMeans algorithm's objective function is:

$$\arg \min \{C_k\}_{k=1}^K \sum_{k=1}^K \sum_{x \in C_k} \|x - \mu_k\|^2 \quad (7)$$

where C_k is the set of features in the k th cluster, μ_k is the mean vector of features in the cluster, or the corresponding prototype center, and K is the number of

prototypes built for each category ($K=3$ in this paper). The KMeans algorithm divides the data into clusters and finds the center of each cluster by summing the squares of the distances from each sample point to the cluster center to which it belongs. This ensures that the samples in the clusters are as close together as possible. It should be noted that the KMeans algorithm serves solely as an offline unsupervised clustering tool to assist in uncovering intra-category diversity. This approach does not interfere with the labeled supervised training process of the main classification task, but rather enhances the robustness and accuracy of distance metric learning and classification by enriching the structural information of prototype centers.

S6.2: Prototype Normalization and Minimum-Distance Selection

After constructing the prototypes, we perform L2 normalization on all prototypes to eliminate the effect of feature scale differences on subsequent distance calculations. For each prototype μ_k , the normalized prototype is represented as follows:

$$\hat{\mu}_k = \frac{\mu_k}{\|\mu_k\|_2} \quad (8)$$

Subsequently, when classifying samples from a Query set, we compute the Euclidean distance between each query sample feature and multiple prototypes of the corresponding category separately, and select the smallest of these distances as the distance between that query sample and that category:

$$d(x_q, C) = \min_{k=1, \dots, K} \|x - \hat{\mu}_k\|_2 \quad (9)$$

where x_q is the features of the query sample, C is the set of prototypes in a category.

This minimum distance strategy can effectively capture the nearest local features

within a category, thus improving the accuracy of classification.

S7 Appendix. Loss Function Design

S7.1 Multi-prototype Classification Loss

In order to overcome the shortcoming of traditional prototype networks that cannot adequately express the diversity within the categories with only a single center, we generate $K=3$ prototypes for each category by KMeans on the support set $\{\mu_{c,k}\}_{k=1}^3$ and adopt the “minimum distance” criterion in the query stage for classification. Specifically, let the query sample be characterized as x_q , and its distance to category c is defined as:

$$d_c(x_q) = \min_{1 \leq k \leq 3} \|x_q - \mu_{c,k}\|_2 \quad (10)$$

and use negative distances as categorical logits:

$$l_c(x_q) = -d_c(x_q) \quad (11)$$

Weighted cross-entropy was used for the final loss:

$$L_{\text{proto}} = - \sum_{c=1}^C \omega_c 1\{y_q=c\} \left[l_c(x_q) - \log \sum_{j=1}^C e^{l_j(x_q)} \right] \quad (12)$$

where ω_c is set based on category frequency to mitigate positive and negative category imbalances.

In principle, the multi-prototype approach is equivalent to a multi-peak (Mixture) approximation of the class-conditional distribution $p(x|y=c)$ for each category, i.e., each prototype $\mu_{c,k}$ corresponds to a high-density subgroup. Selecting the center of the nearest subpopulation when querying is able to capture the multimodal structure

within the category, which in turn forms a segmented decision boundary consisting of Voronoi regions; at the same time, using the minimum distance instead of the mean distance effectively reduces the sensitivity to outliers and noise.

S7.2 Anchor Alignment Loss

Although `AnchorFeatureGenerator` is capable of generating diverse pseudo-samples, if its features are semantically offset from the true category center (Anchor), it introduces noise and weakens subsequent prototype learning. For this reason, we introduce an Anchor alignment loss in training by constraining the cosine similarity of each pseudo-sample feature f_i to its corresponding SwAV pre-trained Anchor vector a_{y_i} :

$$L_{\text{align}} = 1 - \frac{1}{N} \sum_{i=1}^N \frac{f_i \cdot a_{y_i}}{\|f_i\|_2 \|a_{y_i}\|_2} \quad (13)$$

where N is the total number of pseudo-samples and label y_i corresponds to the center of the category in which they should be clustered. This alignment strategy imposes semantic consistency constraints on the generated samples directly in the feature space, which significantly reduces the drift between pseudo-samples and the true distribution and improves the validity of pseudo-samples in subsequent prototype construction. Compared to adversarial domain-alignment methods, our approach is computationally simpler, integrates seamlessly with multi-prototype classification, and directly preserves angular feature relationships via cosine-similarity optimization, thereby enhancing the discriminative power of the embeddings.

S7.3 Batch-Hard Triplet Loss

To further tighten the category boundaries of the feature space, we use the "Batch-Hard" triad loss, which automatically picks only the most discriminative positive/negative pairings of samples inside each mini_batch for optimization. First, we calculate the Euclidean distance matrix ($D_{ij}=\|e_i-e_j\|_2$) between all embedding vectors inside a batch, and then for each sample, we select the "hardest positive sample (d_i^{pos})" with the biggest distance among the same-labeled samples i , and the "hardest negative sample (d_i^{neg})" and:

$$L_{\text{triplet}} = \frac{1}{B} \sum_{i=1}^B \max(d_i^{\text{pos}} - d_i^{\text{neg}} + m, 0) \quad (14)$$

Optimization (bounded hyperparameter, $m=0.5$) Compared to random sampling, Batch-Hard focuses only on the most difficult pairs of samples, which not only avoids the redundant gradient caused by easy-to-classify samples, but also significantly accelerates model convergence and strengthens discriminative ability in the case of fewer samples, particularly when negative samples are scarce and unevenly distributed. This module, along with multi-prototype categorization and Anchor alignment loss, creates a more robust and expressive embedding space.

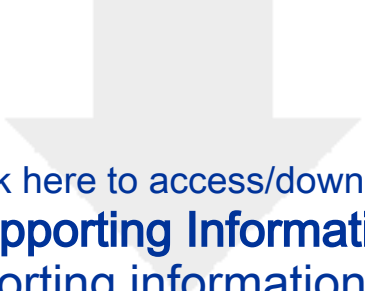
S7.4 Weighted Combination of Overall Loss

Finally, we weighted and combined the three sub-losses according to the following formula to form the ultimate training objective:

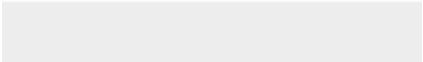
$$L = L_{\text{proto}} + \alpha L_{\text{align}} + \beta L_{\text{triplet}} \quad (15)$$

The weighting design ($\alpha=0.2$, $\beta=0.5$) ensures the multi-prototype classification loss dominates the main task while balancing the constraints of the Anchor alignment

1080 loss on pseudo-sample semantic consistency and the Batch-Hard triplet loss's
1081 reinforcement of embedding space boundaries through smaller weights. This scheme
1082 references Kim et al.'s weight allocation method for sample fusion in
1083 class-imbalanced tasks while incorporating Li et al.'s loss balancing experience from
1084 tabular data augmentation scenarios. Through this synergistic multi-objective loss
1085 optimization, the current framework achieves accurate classification decisions in
1086 small-sample multimodal scenarios while maintaining clear feature space organization,
1087 providing robust support for subsequent model generalization and robustness.



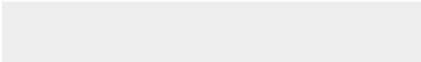
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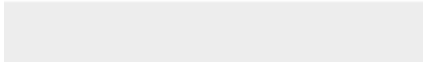




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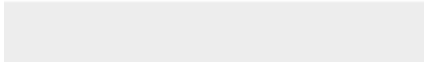




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Few-Shot Prediction of Post-Fusion Acute Kidney Injury Using Multimodal Radiomics-Clinical Feature-Based Prototype Networks

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Abstracts

To optimize preoperative decision-making, the prediction of acute kidney injury (AKI)—a major postoperative complication—requires early identification. This study developed a multimodal few-shot prototype network based on real-world clinical data from 194 single-center cases. The model incorporated 11 preoperative parameters encoded by a multilayer perceptron (MLP) to weight and integrate imaging features extracted by ResNet-50, with a fixed loss function implemented to address class imbalance. Results demonstrated that the fused model achieved an area under the curve (AUC) of 0.816, significantly outperforming the clinical-only (AUC = 0.578) and imaging-only (AUC = 0.510) models. Interpretability analysis using SHapley Additive exPlanations (SHAP) revealed the influence of key variables (age, lactate, creatinine). Our study demonstrates that multimodal fusion significantly improves generalization performance in data-limited scenarios, establishing the first application of few-shot learning with prototype networks for postoperative AKI prediction. We recommend incorporating 11 standardized preoperative indicators and imaging data into clinical risk assessment to enhance predictive accuracy and utility. Future directions should focus on expanding cohort sizes and refining radiomic feature extraction.

Keywords: Acute kidney injury (AKI), Few-shot learning, Prototypical networks, Multimodal fusion, SHAP value analysis

1. Introduction

Acute kidney injury (AKI) is a clinically critical condition characterized by an acute decline in renal function with concomitant disturbances in fluid, electrolyte, and acid-base homeostasis. **The Kidney Disease: Improving Global Outcomes (KDIGO) criteria define AKI as:** 1. Serum creatinine (SCr) rises by $\geq 26.5 \mu\text{mol/L}$ within 48 hours; or 2. SCr increases to >1.5 times baseline within 7 days; or 3. Urine output remains $<0.5 \text{ mL}/(\text{kg}\cdot\text{h})$ for 6 hours[1].

Postoperative acute kidney injury (AKI) is prevalent among hospitalized patients, particularly following major surgeries. Its occurrence correlates strongly with increased mortality, prolonged hospitalization, elevated risks of chronic kidney disease progression, and higher healthcare costs[2][3]. In high-risk surgical populations, the incidence of acute kidney injury (AKI) is reported to exceed 20%, and it is notably higher in intensive care unit (ICU) patients[3]. The clinical utility of current diagnostic markers is limited by their substantial time lag, which often fails to detect reversible injury during the early stages of AKI, as highlighted by previous studies[2],[4]. Therefore, early identification and risk stratification are critical for implementing effective renal protection strategies.

As artificial intelligence has advanced, an increasing number of studies have attempted to use deep learning or machine learning techniques for preoperative AKI prediction tasks. Most of these models build risk classifiers using algorithms such as logistic regression, random forests, or multilayer perceptrons, with structured

63 electronic health record (EHR) data as input. However, the accuracy and
64 generalization ability of these models are often limited by traditional approaches,
65 which tend to rely heavily on tabular features while neglecting unstructured data, such
66 as changes in patient images and physiological variations[5]. This leads to prediction
67 bias and performance imbalances, as traditional models are more likely to predict the
68 majority class during training. Postoperative AKI, a low-incidence but high-risk event,
69 is particularly affected by this issue due to the significant imbalance between positive
70 and negative samples, especially in real-world hospitalized populations. In China,
71 acute kidney injury (AKI) has a high incidence among hospitalized patients, with an
72 estimated prevalence of 20% [6].

73 Furthermore, most current models lack clinical interpretability. In real-world
74 medical practice, clinicians often require clear explanations of a model's rationale
75 before trusting algorithm-assisted judgments. Conventional "black-box" AI models
76 typically provide insufficient decision transparency, which limits their adoption and
77 deployment in clinical settings[7].

78 Additionally, unbalanced data remains a major challenge in postoperative AKI
79 risk prediction, as the number of AKI cases often exceeds that of non-AKI patients.
80 This imbalance leads to overfitting and misclassification in traditional machine
81 learning algorithms, which favor the majority class. In clinical settings, such
82 imbalance can significantly undermine model stability and dependability.
83 Furthermore, AKI is a multifactorial condition that involves complex interactions

between factors such as anesthetic procedures, intraoperative blood loss, postoperative infections, and electrolyte imbalances. Therefore, a more comprehensive approach integrating postoperative medical imaging with tabular clinical data is essential for accurate risk identification and etiological modeling.

Few-shot learning (FSL) has increasingly drawn attention in this context. By enabling classifiers to generalize well from only a handful of labeled examples, FSL addresses both the high cost of annotation and the scarcity of medical data. In particular, metric-learning frameworks such as Prototypical Networks (PNs) infer the class distribution of novel samples from a small support set, making them especially well suited to low-incidence, high-risk clinical prediction tasks like postoperative AKI.

In this study, a multimodal small-sample prototype network framework combining preoperative medical imaging and structured form data was built for the problem of postoperative AKI prediction using actual clinical data from **a single center (the Second People's Hospital of Sichuan Province)**. A separate multilayer perceptron (MLP) with weighted fusion encodes the structured form data, while a pre-trained Residual Network (ResNet-50) model extracts features from the image data. We describe the You Only Look Once (YOLO) labeled region cropping mechanism to enhance the efficacy of picture representation by improving the efficacy of image features. associated loss function was created to enhance the model's recognition capabilities on a limited number of sample classes in order to

Commented [1]: Reviewer's comments: The paper only presents incidence rates and adverse effects, but fails to explain “why laparoscopic nephrectomy was selected as the scenario” or clarify “the unique significance of AKI prediction in this scenario and the research gap.”

Supplementary explanation and revision:

In recent years, minimally invasive surgical techniques, particularly robot-assisted partial nephrectomy (RAPN) and laparoscopic partial nephrectomy (LPN), have gained widespread global adoption. Compared with open surgery, these minimally invasive approaches achieve comparable oncologic outcomes while offering advantages such as fewer complications and faster recovery[a]. Long-term follow-up data indicate no significant differences in cancer control and renal function preservation between LPN and open partial nephrectomy (OPN)[b]. Despite the availability of robotic-assisted surgery in large medical centers, the majority of hospitals in China still rely primarily on laparoscopic techniques for both partial and radical nephrectomies.

[a]Kawase K, Enomoto T, Kawase M, et al. The Impact of Postoperative Renal Function Recovery after Laparoscopic and Robot-Assisted Partial Nephrectomy in Patients with Renal Cell Carcinoma. *Medicina*. 2022;58(4):485. doi:10.3390/medicina58040485.

[b]Nicaise E, Feldman AS, Gusev A, et al. A contemporary comparison of laparoscopic versus open partial nephrectomy for renal cell carcinoma. *BMC Urol*. 2024;24(1):58. doi:10.1186/s12894-024-01423-w.

We focus on acute kidney injury (AKI) following laparoscopic renal surgery (partial or radical nephrectomy) because this procedure entails dual renal stress mechanisms—reduced renal perfusion induced by pneumoperitoneum and ischemia–reperfusion injury—making patients more susceptible to AKI and conferring greater renal vulnerability compared with other abdominal or non-abdominal surgeries.

According to the KDIGO criteria, the incidence of AKI after partial nephrectomy ranges from 15% to 25% **【1,2】**, and postoperative AKI is strongly associated with long-term renal function decline and an increased risk of new-onset

...

address the issue of data imbalance. In addition to increasing the model's accuracy in identifying postoperative AKI risk, this study offers a workable route and theoretical foundation for future high-dimensional medical multimodal modeling and illness prediction on a small number of samples.

In summary, this study has contributed to the predictive modeling of postoperative acute kidney injury (AKI) in a number of noteworthy and creative ways, including:

1. Our study addresses a critical gap in AKI prediction research—the limited diagnostic dimensions—by combining clinical indicators and imaging data with strict adherence to KDIGO diagnostic criteria.

2. We focus on AKI after laparoscopic renal surgery (partial resection/nephrectomy), which has been reported to occur in 9–25% of cases[8][9] and is associated with an increased risk of postoperative dialysis, as reported by various studies[1]. To address this, we developed a surgery-specific prediction model.

3. Our few-shot prototype network integrates multimodal data (preoperative static variables, early physiological trends, and imaging), significantly improving early-warning performance in limited-data scenarios.

4. Model development used **real-world, single-center data from Sichuan Provincial Second People's Hospital**, addressing regional gaps in generic AKI models by incorporating local epidemiological, environmental, and healthcare system factors.

5. SHAP interpretability analysis enhances clinical utility by visualizing risk factors,

enabling physicians to understand model predictions and improve real-world adoption of this intelligent system.

In conclusion, this study offers a theoretical foundation and workable technical support for the early detection and tailored treatment of postoperative AKI in resource-constrained areas through the development of a practical, interpretable, and geographically flexible postoperative AKI prediction framework.

2. Theoretical Basis and Research Progress

2.1. Advances in Postoperative AKI Prediction Research

Research in clinical renal protection management has focused on the prediction and early detection of postoperative acute kidney damage (AKI), one of the frequent consequences in the perioperative period. The AKI-EPI study provides a comprehensive analysis of the epidemiology of AKI in critically ill patients across multiple centers[10]. Based on imaging histological findings and structured clinical variables, AKI prediction algorithms may currently be generally divided into two groups.

Clinical feature-based prediction techniques typically rely on conventional machine learning models, such as logistic regression or random forests, which utilize basic preoperative data, intraoperative physiological markers, and perioperative laboratory results. For instance, Liu et al. developed a logistic regression model to predict AKI following liver transplantation using preoperative age, blood pressure, and creatinine levels, achieving an AUC of 0.78 on the validation set, demonstrating

acceptable clinical feasibility[11]. Rank et al. (2020) reported that a random forest model achieved the highest performance in predicting post-cardiac surgery AKI, with an area under the curve (AUC) of 0.88 and a sensitivity of 0.82[12]. Although these methods are straightforward to implement and highly interpretable, they struggle to capture complex nonlinear relationships within the data and thus often have limited predictive accuracy, particularly when variable interactions are significant or when manually constructing higher-order features is difficult.

The main goals of imaging histology techniques are to identify patients' possible risk of renal damage using multidimensional vectors and to extract deep quantitative information from medical pictures (such as postoperative CT, ultrasound, etc.). These techniques can improve the ability to measure the risk of AKI in preclinical stages and aid in the clinical detection of "non-visible" early pathological changes on imaging[13]. Zhou et al. developed a radiomics model to predict AKI following heart surgery by extracting renal texture features from postoperative abdominal CT images. The model demonstrated good image-assisted discrimination, achieving an AUC of 0.81 in external validation[14]. However, these techniques are subject to specific technical thresholds in their widespread promotion and use because to their heavy reliance on image quality, segmentation accuracy, and annotation consistency, as well as the absence of consistent standards in multi-center settings.

Current methods for predicting postoperative AKI still encounter several challenges during model development, including small overall sample sizes, class

imbalance, and limited multimodal feature–fusion capabilities. Although many studies report a pronounced imbalance between positive and negative cases, the more fundamental issue is the insufficient total number of samples. This scarcity makes training vulnerable to small-sample effects, leading to unstable parameter estimates and reduced generalizability. In particular, even when class ratios are not severely skewed, underrepresentation of minority classes (e.g., non-AKI) can impair model reliability and sensitivity in clinical applications. To address these issues—small samples, class imbalance, and weak multimodal fusion—we propose a prototypical-network approach that integrates postoperative imaging histology with clinical features within a few-shot learning framework, thereby enhancing early detection performance and model robustness for postoperative AKI.

2.2. Multimodal Radiomics Feature Fusion

Recent years have seen a large application of multimodal feature fusion techniques in the field of medical risk prediction, particularly for tasks with high clinical importance like kidney lesions and surgical sequelae. Radiomics is a technique that helps with risk stratification and disease prediction judgment by extracting high-dimensional and quantitative features from medical images (such as CT, MRI, and ultrasound) that can detect early pathological changes that are not detectable by traditional visual assessment. Gillies et al. introduced the concept of radiomics, emphasizing the extraction of quantitative features from medical images for decision support[15]. Information like structural alterations, perfusion status, and

kidney interstitial features are frequently seen in postoperative images as slight texture or density changes. Imaging histology may be able to serve as a foundation for identifying these asymptomatic phases, which is important for the prediction of postoperative AKI.

In an effort to provide a more comprehensive risk prediction framework, increasing numbers of studies have attempted to simultaneously model radiomic features with structured clinical characteristics (such as blood creatinine, lactate, and urine output changes). Additionally, some research has investigated fusion architectures based on dual-branch neural networks for imaging and tabular data, which enhance the model's ability to incorporate cross-modal heterogeneous data through adaptive weighting and attention mechanisms[16].

Although current research has preliminarily verified the potential of multimodal fusion strategies for complex prediction tasks such as postoperative AKI, several practical challenges remain: First, dimensional disparities and inconsistent expression mechanisms across modalities often lead to semantic misalignment and information redundancy during fusion. Second, non-standardized medical imaging acquisition protocols frequently introduce image quality variations and segmentation errors that compromise feature stability. Finally, the complex architectures of multimodal models typically demand large training datasets and high annotation precision, limiting their generalizability in small-sample scenarios. To address these limitations, future research should focus on developing efficient, generalizable lightweight fusion

strategies to enhance model stability and cross-scenario adaptability.

2.3. Few-Shot Learning Methods

In recent years, few-shot learning (FSL) has gained popularity in medical image analysis. Its goal is to enable effective modeling with very few labeled samples. Few-shot learning is particularly valuable when medical image data acquisition is challenging and manual annotation is costly. The primary objective is to achieve rapid generalization and accurate discrimination of new sample categories by constructing metric spaces or refining meta-learning strategies using limited support samples[17].

Prototypical Networks (PN), a widely used metric learning approach in FSL, classify samples by calculating feature means of support samples as class prototypes. Query samples are then compared to these prototypes using Euclidean distance. This architecture has demonstrated significant success in various medical applications including breast cancer screening[18], brain tumor classification[19], and fundus lesion detection[20], owing to its simplicity, training stability, and strong generalization capability under data scarcity.

Furthermore, Matching Networks[21] enhance small-sample structural discrimination by considering global matching relationships between query and support samples, while Relation Networks[22] improve fine-grained lesion recognition through learned nonlinear similarity functions. Recent studies have also integrated FSL with multimodal fusion techniques, such as jointly embedding structured clinical data and imaging features into a shared representation space[23], to

231 boost prediction accuracy via collaborative modeling. These fusion strategies have
232 progressively expanded FSL's clinical applications in areas like diagnostic assistance
233 and postoperative risk prediction.

234 3. Methodology

235 3.1. Study Subjects and Data Collection

236 3.1.1. Participant Eligibility Criteria

237 Archived clinical and imaging data were accessed for this retrospective study on
238 1 May 2025. No identifiable data were accessed, and the authors did not have access
239 to any information that could identify individual participants during or after data
240 collection.

241 According to the Kidney Disease: Improving Global Outcomes (KDIGO) criteria,
242 postoperative acute kidney injury (AKI) is defined as: (1) an increase in serum
243 creatinine (SCr) by $\geq 26.5 \mu\text{mol/L}$ within 48 hours; (2) SCr rising to >1.5 times the
244 baseline value within 7 days; or (3) urine output $<0.5 \text{ mL}/(\text{kg}\cdot\text{h})$ for more than 6
245 hours. In this study, the lowest SCr value within 6 months prior to surgery served as
246 the baseline creatinine level for postoperative AKI assessment.

247 This retrospective study analyzed patients with renal space-occupying lesions
248 who underwent laparoscopic renal surgery at **Sichuan Provincial Second People's**
249 **Hospital**. All participants were selected through strict inclusion and exclusion criteria
250 to ensure population homogeneity and minimize confounding factor interference.

251 **Inclusion Criteria:**

Commented [2]: Merge with the original third paragraph

Final version:

This retrospective study analyzed patients with renal space-occupying lesions who underwent laparoscopic renal surgery at **Sichuan Provincial Second People's Hospital**. Archived clinical and imaging data were accessed on 1 May 2025. All participants were selected according to strict inclusion and exclusion criteria to ensure population homogeneity and minimize confounding factors. No identifiable data were accessed during or after data collection, and the authors had no access to any information that could identify individual participants. All data were anonymized to ensure confidentiality.

Commented [3]: Remove duplicate definitions

Commented [4]: Merge with the first paragraph of the original text and provide a brief summary.

(1) Patients with radiologically confirmed renal space-occupying lesions (via ultrasound, urological CT, or intravenous urography (IVU)).

(2) Patients who underwent either laparoscopic partial nephrectomy or radical nephrectomy at our institution, with all surgeries performed by the same experienced chief urologist.

(3) Adult patients aged > 18 years.

(4) Patients who had serum creatinine levels measured at least once preoperatively and within 7 days postoperatively.

(5) Patients with complete baseline clinical data.

Exclusion Criteria:

(1) Patients with end-stage renal disease (ESRD).

(2) Patients undergoing concomitant surgeries.

(3) Patients with a history of renal transplantation.

(4) Patients with other major organ diseases, malignancies, or undergoing chemotherapy.

(5) Patients missing critical data.

This study might more accurately determine the incidence of acute renal damage following surgery while accounting for the impact of significant confounding variables by screening the study participants using the aforementioned criteria. The procedure for selecting study participants based on predetermined inclusion criteria is shown in **Fig 1**.

3.1.2. Multimodal Data Acquisition

Numerous machine learning and deep learning models using preoperative clinical features have been developed to predict postoperative acute kidney injury (AKI), but these unimodal approaches still have limitations. First, most studies rely solely on clinical variables—such as biochemical blood indices, vital signs, and medical history—to construct risk scores or train models. For example, Yoon et al.’s CMC-AKIX model, built on 38 preoperative clinical variables, achieved an AUC of 0.83 in a retrospective cohort; however, its applicability is largely confined to specific surgical types and requires larger-scale external validation before it can be generalized to wider clinical settings[24]. Furthermore, a systematic evaluation of 150 AKI prediction models highlighted substantial variability in study design, cohort composition, AKI definitions, and predictor variables, leading to high risk of bias and heterogeneity, which limit the models’ clinical utility[25]. More importantly, clinical data alone may fail to capture microanatomical or functional changes apparent in preoperative imaging, which often manifest before biochemical markers such as serum creatinine become abnormal.

To address these limitations, this study integrates two complementary data types—preoperative clinical indicators and medical imaging—through a multimodal learning framework. Multimodal learning, a machine learning paradigm that combines heterogeneous inputs from diverse sources (e.g., images, text, tabular data), enhances model performance by exploiting inter-modality complementarity[26].

294 Specifically, we propose and quantify an imaging-derived anatomical parameter
295 ("**solitary kidney status**") and integrate it with clinical indicators in our predictive
296 model.

297 The imaging-derived feature "isolated kidney status" holds significant clinical
298 importance. A retrospective study demonstrated that patients with congenital or
299 surgically acquired solitary kidneys faced substantially higher AKI risks during
300 procedures with potential renal injury, such as percutaneous angiography. In a cohort
301 of 3,456 angiography patients, Mayo et al. reported an odds ratio (OR) of 1.9 (95% CI
302 1.3-2.8) for AKI occurrence in the solitary kidney group versus the two-kidney
303 group[27]. Additionally, Wang et al.'s cohort study of 786 renal tumor patients
304 revealed a postoperative AKI incidence of 24.7% in patients with preoperative
305 unilateral nephrectomy history, significantly higher than the 12.5% incidence in
306 patients with two kidneys ($P<0.01$)[28]. These findings confirm that solitary kidney
307 status effectively reflects interindividual variability in renal functional reserve.
308 Incorporating this parameter into postoperative AKI prediction models could enhance
309 high-risk patient identification accuracy and guide personalized preoperative
310 interventions.

311 This study developed a multimodal fusion strategy for postoperative AKI risk
312 prediction by integrating feature representations from two complementary data
313 modalities: medical imaging and clinical indicators. For the clinical modality, we
314 incorporated significant preoperative clinical parameters (blood glucose, creatinine,

lactate, and urea nitrogen) based on international guidelines, along with the imaging-derived parameter "**solitary kidney status.**" For the imaging modality, we employed a pre-trained ResNet50 network to automatically extract deep spatial features from preoperative images, including the critical "solitary kidney status" assessment. To comprehensively capture patients' anatomical and physiological status, we performed cross-modal fusion of these features using both feature concatenation and a gated weighting module, generating a unified multimodal representation. Our data integration approach not only addresses limitations of unimodal analysis but also establishes a robust foundation for postoperative AKI prediction.

3.1.3. Clinical Feature Selection and Definition

This study utilized several preoperative clinical indicators to develop a predictive model for postoperative acute kidney injury (AKI) in patients undergoing renal laparoscopic surgery. These indicators provide a comprehensive assessment of patients' hemodynamic status, metabolic abnormalities, and renal function, offering critical information for postoperative AKI risk prediction.

To provide a more comprehensive assessment of patients' baseline renal function and hemodynamic status, the model initially incorporated traditional preoperative markers including **age, serum creatinine, urea nitrogen, and hemoglobin**. Older patients show increased vulnerability to acute kidney injury (AKI) due to age-related renal function decline. Elevated serum creatinine and urea nitrogen levels typically indicate impaired renal function and consequently higher AKI risk. Hemoglobin

levels reflect the patient's anemia status, which may lead to postoperative renal ischemia and consequently increase AKI incidence.

To further enhance the model's sensitivity to preoperative metabolic disorders and tissue perfusion changes, this study innovatively introduced two additional indicators: **preoperative glucose and lactate levels**. Preoperative glucose levels have demonstrated a close association with AKI risk, showing approximately 15% increased risk for every 1 mmol/L elevation in non-cardiac surgical patients[29]. Similarly, preoperative lactate levels are strongly associated with severe AKI occurrence, as studies have shown that postoperative lactate levels >3.3 mmol/L at 12 hours significantly increase severe AKI risk[30].

Two renal function-associated risk factors—**hypertension** and **diabetes mellitus**—were also included in the analysis. Diabetes mellitus induces renal damage through altered glucose metabolism and is associated with higher postoperative AKI incidence, especially in patients with inadequate glycemic control. Hypertension, defined as systolic blood pressure ≥ 140 mmHg or diastolic blood pressure ≥ 90 mmHg, is an established risk factor for chronic kidney disease.

Furthermore, **blood urea nitrogen (BUN)** and **erythrocyte specific volume** were selected as traditional indicators of renal function and hemodynamics for this model. Low erythrocyte specific volume or perioperative anemia significantly elevates postoperative AKI incidence by reducing tissue oxygen supply, particularly during major surgeries with substantial blood loss, rendering anemic patients more

vulnerable to AKI. BUN serves as a crucial glomerular filtration rate marker and, together with serum creatinine, represents a standard monitoring parameter for AKI diagnosis and management[31].

3.2. Data Preprocessing

To preserve sample size and minimize extreme value effects, missing preoperative clinical values were replaced with column-wise medians. Comparative studies on medical datasets confirm that median imputation achieves predictive performance comparable to complex multiple imputation methods[32]. Crucially, the prototype network's episode training—which dynamically reorganizes the support set to regularly expose the model to new sample combinations—effectively mitigates overfitting from fixed imputation values. For corrupted or unloadable images, full grayscale mapping ensures multimodal input continuity and processing robustness.

After imputing missing values in predefined clinical variables using median imputation, we applied scikit-learn's StandardScaler to compute the mean and standard deviation of each variable on the training set, transforming all features to a standard normal distribution (zero mean and unit variance). This procedure removes scale disparities, prevents large-magnitude features from dominating gradient updates, and accelerates neural network convergence while enhancing stability and generalization. Proper feature normalization has been shown to substantially affect classification model performance[33]. To ensure pipeline reproducibility, we serialized and saved the StandardScaler parameters to apply identical normalization to

Commented [5]: 3.2 Data Preprocessing and 3.3 Data Augmentation are combined to highlight the core technology while removing detailed descriptions.

After modification:

3.2 Data Preprocessing and Augmentation

To preserve sample size and minimize the influence of outliers, missing preoperative clinical values were replaced with the median of each variable. Prior studies on medical datasets have demonstrated that median imputation yields predictive performance comparable to more complex multiple-imputation techniques [32]. To remove unit disparities and facilitate model convergence, Z-score normalization was subsequently applied to all continuous variables [33]. The complete imputation and normalization workflow is provided in Methods S1.

In the image pipeline, regions of interest (ROIs) were cropped according to YOLO annotations and rescaled to a uniform resolution. Lightweight geometric and luminance augmentations were applied to increase data diversity before feature extraction by a pretrained network (Methods S2). Given the pronounced class imbalance in the dataset (132 AKI vs. 62 non-AKI cases), a multimodal framework combining feature-space interpolation, weighted sampling, and class-weighted loss was employed to improve model sensitivity to the minority class.

Furthermore, an anchor-based pseudo-sample generation strategy was proposed to further expand the negative-class feature space while maintaining multimodal semantic consistency (Fig. 2). This approach utilizes pretrained feature vectors to guide pseudo-sample generation, thereby enhancing data diversity and model interpretability. The parameter configurations and algorithmic details are described in Methods S2.

new data during inference, thus completing the end-to-end process from raw clinical records to model inputs.

Commented [6]: Technical details have been moved to Supporting information S1 Appendix. Data imputation and standardization.

After median imputation, predefined clinical fields are extracted. Using scikit-learn's StandardScaler, the mean and standard deviation of each variable are calculated on the training set, transforming all features into a standard normal distribution with zero mean and unit variance. This normalization eliminates magnitude differences, prevents large variables from dominating gradient updates, accelerates neural network convergence, and improves stability and generalization performance. For the image processing branch, regions of interest (ROIs) are first cropped based on YOLO annotations and rescaled to original pixel coordinates. The images then undergo augmentation including random minor rotations, online horizontal flipping, brightness/contrast adjustment, and resampling to 224×224 resolution. A pretrained ResNet50 model (with final classification layer removed) subsequently extracts 2048-dimensional deep features for multimodal fusion.

3.3. Data Augmentation and Class Imbalance Handling

Commented [7]: Has been incorporated into the original Section 3.2.

To address the highly imbalanced distribution of positive (132 AKI) and negative (62 non-AKI) cases in our dataset, we developed two augmentation strategies for image and tabular modalities within our multimodal framework, alongside resampling. In the image branch, regions of interest were first cropped, then subjected to lightweight geometric and luminance transformations (random rotations of $\pm 5^\circ$, brightness/contrast adjustments of $\pm 5\%$, and random horizontal flips), followed by

resizing to 224×224 pixels. In the tabular branch, we adopted a SMOTE-inspired approach: positive-class samples were randomly paired and linearly interpolated over the $[0, 1]$ interval to generate synthetic examples, while negative-class samples were interpolated along the prototype network’s negative-sample boundary to create harder negatives. During training, a weighted random sampler and weighted loss function were applied synchronously to increase sensitivity to the minority AKI class and enhance overall model robustness. These three techniques work in concert to balance minority and majority class distributions.

Traditional generative approaches in feature space, such as GANs or Mixup, often overlook semantic consistency across modalities, resulting in samples of unstable quality and poor interpretability[34]. To address this, we propose a three-step enhancement strategy (**Fig 2**). First, diversified pseudo-samples are generated by the AnchorFeatureGenerator from SwAV-pretrained Anchor vectors and random noise to broaden the global distribution of the negative class. Second, we compute Euclidean distances among standardized negative-class tabular features and use the 25th percentile as an adaptive threshold, then perform linear interpolation on nearest-neighbor sample pairs to enrich local distribution diversity. Finally, these pseudo-samples are combined with real negative samples in the support set for multi-prototype construction and loss computation. This strategy enhances interpretability and more comprehensively covers the negative-class feature space.

3.4. Multimodal Feature Fusion Network

Commented [8]: Details of this nature have been omitted from the main text and relocated to the corresponding S2 Appendix. Data Augmentation, Class Balancing, and Feature Extraction

In this study, we propose an innovative multimodal feature fusion framework to fully exploit complementary information from imaging and tabular data in small-sample clinical settings. First, 2048-dimensional **visual features** are extracted from the image branch using a pretrained **ResNet-50**, and 128-dimensional structured features are obtained from the **tabular branch** via a multilayer perceptron encoder and concatenated into a joint representation. Next, parallel sigmoid gating dynamically assigns weights to each modality, amplifying reliable signals while suppressing noise. The weighted high-dimensional vectors are then passed through a two-layer fully connected network incorporating batch normalization, **LeakyReLU activation**, and **dropout**, compressing them into 128-dimensional embeddings. Finally, **L2 normalization** ensures compatibility with metric learning. This lightweight design, which avoids complex attention mechanisms and extensive hyperparameter tuning, flexibly balances each modality's contribution and effectively enhances sensitivity and generalization in imbalanced postoperative AKI prediction tasks.

3.4.1. Design Rationale and Architecture

Imaging and tabulated preoperative measurements can be used to fully utilize the complimentary information of both data sources in the prediction of clinical postoperative acute kidney damage (AKI). However, this multimodal fusion also simultaneously confronts the following significant obstacles in small-sample scenarios:

Commented [9]: The content of the original Section 3.4.1 has been incorporated into the existing text, and the original Section 3.4.1 has been removed.

Revised:

This study presents a multimodal feature fusion framework to effectively leverage complementary information from imaging and tabular data in small-sample clinical contexts, addressing challenges such as data scarcity, modality imbalance, and cross-modal alignment. As shown in Fig. 3, visual features are extracted from the image branch using a pretrained ResNet-50, yielding 2048-dimensional representations, while structured features are obtained from the tabular branch via a multilayer perceptron encoder, resulting in 128-dimensional representations. These features are combined into a joint representation. Next, parallel sigmoid gating assigns dynamic weights to each modality, amplifying reliable signals and suppressing noise. The weighted high-dimensional vectors are compressed into 128-dimensional embeddings through a two-layer fully connected network with batch normalization, LeakyReLU activation, and dropout. Finally, L2 normalization ensures compatibility with metric learning. In imbalanced postoperative AKI prediction tasks, this lightweight architecture improves sensitivity and generalization while avoiding complex attention mechanisms and extensive hyperparameter tuning.

Commented [10]: Delete Section 3.4.1, summarize briefly, and incorporate it into the above general overview paragraph.

- 441 • **Data Scarcity:** The model’s ability to reliably learn high-dimensional image
442 features is constrained by the extremely small number of labeled samples per
443 class—in this study, only 194 cases in total.
- 444 • **Modality Imbalance:** Substantial differences in dimensionality and distribution
445 exist between imaging and tabular features; simple concatenation often allows
446 one modality to dominate while the other is effectively “drowned out”[35].
- 447 • **Semantic Gap:** Visual features and structured indicators cannot be directly
448 aligned at the semantic level and require effective cross-modal mapping to fully
449 leverage their complementary advantages[36].

450 To address these challenges, we propose a custom multimodal few-shot
451 network—MultiModalFewShotNet—that employs dynamic gating and normalized
452 projection(Fig 3).

453 **Key Modules:**

454 **Image Encoder:** Utilizes a pretrained ResNet-50 backbone with the final fully
455 connected layer removed, outputting 2048-dimensional visual features.

456 **Tabular Encoder:** Processes structured tabular data by encoding input features
457 through a neural network (MLP), producing a low-dimensional representation.

458 **Feature Fusion and Normalization:** After weighted fusion of image and tabular
459 features, batch normalization is applied to ensure consistent feature scales, providing
460 a unified input for subsequent processing.

461 **3.4.2. Feature Extraction Module**

Commented [11]: As already mentioned in the overview section, a brief summary is provided here. Detailed network architecture, parameter configurations, and training strategies are documented in S3 Appendix. Detailed Design of the Feature Extraction Module. Revised:

To integrate multimodal data, we use a dual-branch feature-extraction architecture.

Image feature encoder: A pretrained ResNet-50 with YOLO-based ROI cropping provides high-level visual features while minimizing background interference; the design supports flexible fine-tuning for small-sample settings.

Tabular feature encoder: A shallow MLP maps structured clinical variables into a compact representation with ReLU activations and Dropout for better generalization. Implementation details are provided in S3.

Image Feature Encoder:

We selected ResNet-50 as the visual backbone and optimized its output architecture in the following ways to properly utilize the generic representations acquired from extensive visual datasets:

- **YOLO-based ROI cropping:** To find and crop the region of interest (ROI), we translate normalized bounding-box coordinates from YOLO annotations into pixel coordinates. This eliminates background interference and enhances feature extraction.
- **Feature-extraction optimization:** To avoid the information loss that dimensionality reduction can cause, we eliminated the initial 1,000-way classification head and used global average pooling (GAP) to create 2,048-dimensional feature vectors. This preserved the hierarchical representations of the pretrained model.
- **Flexible fine-tuning strategy:** Because the encoder maintains its full forward inference capacity, we are not constrained by the parameter distribution of the initial classification head and can fine-tune only the final layers or do end-to-end training as necessary.
- **Efficiency of inference:** Our method just needs to load pretrained weights and discard a few parameters, which drastically lowers training overhead while preserving great computing efficiency during inference as compared to retraining a deep convolutional network from scratch.

483 By removing the classification head while retaining the complete 2048-dimensional
484 feature outputs, the image encoder achieves two key advantages: it maximally
485 preserves hierarchical semantic and textural representations learned from large-scale
486 visual datasets, and maintains seamless compatibility with downstream modules. This
487 design eliminates additional parameter training burdens while supporting flexible
488 fine-tuning (either partial or end-to-end) across diverse task scenarios, thereby
489 significantly enhancing the model's adaptability and efficiency for small-sample
490 multimodal fusion tasks without compromising its high-dimensional representational
491 capacity.

492 **Tabular Feature Encoder**

493 We developed a shallow multilayer perceptron (MLP) that is suitably nonlinear
494 for structured preoperative clinical indicators. The following is the design and
495 implementation:

- 496 • **Data preprocessing and missing value handling:** Prior to network input, all
497 preoperative indicators underwent Z-score normalization to standardize
498 distributions across varying scales, preventing learning rate and gradient update
499 imbalances caused by variable scale differences. Missing values were addressed
500 through median imputation to improve model robustness when handling
501 incomplete real-world clinical data.
- 502 • **Two-layer nonlinear mapping:** The original high-dimensional clinical indicators
503 were initially projected into a 128-dimensional space to capture fundamental

correlations, followed by a second linear transformation within the same dimensionality to extract deeper feature interactions. This dual-layer approach enhances feature expressiveness in few-shot learning scenarios.

- **ReLU activation and Dropout regularization:** Each linear transformation incorporated ReLU activation to introduce nonlinearity and alleviate gradient vanishing. Dropout regularization (drop rate=0.2[37]) was subsequently applied to randomly disable neurons, reducing dependence on specific features and substantially boosting few-shot generalization capability.

The tabular encoder, through its "dual-linear transformation + ReLU activation + Dropout regularization" architecture, achieves comprehensive nonlinear abstraction and deep interaction mining of preoperative clinical indicators. This design effectively suppresses overfitting risks in data-limited scenarios while providing stable, high-quality structured feature representations for subsequent multimodal fusion and prototype learning.

3.4.3. Dynamic Gated Fusion Mechanism

To achieve effective integration of visual and structured data under limited-sample constraints, we developed a dynamic gated fusion mechanism that adaptively assigns modality-specific contribution weights according to each sample's feature distribution. This approach maintains robust cross-modal interactions while mitigating information imbalance and overfitting risks through sample-adaptive weight allocation. The complete module architecture operates as follows:

Commented [12]: Combine the original 3.4.3 with the original 3.4.4 to form:

3.4.3 Dynamic Gated Fusion and Dimensionality Reduction

To fuse modalities, we adopt **sample-wise sigmoid gating** that independently weights each modality, avoiding the sum-to-one competition of softmax-based attention. Informative modalities can thus be jointly amplified, while unreliable inputs are automatically down-weighted. The gated representation is then passed through a **lightweight two-layer projection** to obtain a **compact 128-dimensional embedding**, followed by **L2 normalization** to map embeddings onto the unit sphere and harmonize Euclidean and cosine metrics. In Prototypical Networks, class prototypes are obtained by calculating the simple mean of normalized embedded vectors from the support set, and the normalized prototypes also lie on the unit sphere, ensuring consistency and interpretability in distance metrics between classes [38]. This design requires no extra hyperparameter tuning and improves sensitivity and generalization under small-sample and class-imbalanced conditions (Detailed equations and network parameters are provided in Supplementary Material S4.S5)

Feature splicing: the 2048-dimensional visual features(f_{img}) output from the image encoder are spliced with the 128-dimensional structured features output from the table encoder(f_{tab}) in the feature dimension to obtain a (2048+128)-dimensional joint vector:

$$f_{concat} = [f_{img}; f_{tab}] \in \mathbb{R}^{2176} \quad (1)$$

- **Two-layer fully-connected:** the spliced vectors are sequentially passed through a two-layer fully-connected network, each layer coupled with ReLU activation, to capture higher-order interactions across modalities:

$$h_1 = \text{ReLU}(W_1 \cdot f_{concat} + b_1), \quad h_2 = \text{ReLU}(W_2 \cdot h_1 + b_2) \quad (2)$$

In the formula: $W_1 \in \mathbb{R}^{128 \times 2176}, W_2 \in \mathbb{R}^{2 \times 128}$

- **Sigmoid output dual-channel weighting:** replacing the traditional softmax attention mechanism with independent sigmoid gating, independently weighted multimodal feature fusion, applying sigmoid directly to the two output channels:

$$[g_{img}, g_{tab}] = \sigma(W_3 \cdot h_2 + b_3), \quad g_{img}, g_{tab} \in (0,1) \quad (3)$$

Where g_{img} is used to weight visual features and g_{tab} is used to weight structured features.

Conventional multimodal fusion techniques, such as Softmax fusion or multi-head attention, require the weights across all channels to sum to 1. When one modality contributes less, its weight becomes excessively suppressed, making valuable information harder to utilize. In contrast, our approach employs Sigmoid gating to independently weight each modality channel while dynamically adjusting the fusion ratio based on sample characteristics, thereby removing the sum-to-1

Commented [13]: Specific formula details are moved to S4 Appendix. Dynamic Gated Fusion Mechanism

constraint. When both modalities are important, their contributions are simultaneously amplified; when one modality is unreliable, its Sigmoid output approaches 0 to automatically suppress noise. This strategy requires no additional hyperparameter tuning, provides flexibility for handling small sample sizes or outlier data, and enhances the model's focus on the most diagnostically relevant features to improve predictive performance.

3.4.4. Multimodal Feature Fusion and Dimensionality Reduction

After obtaining the adaptive gating weights for the image and tabular branches, the next step is to reconstruct and fuse the two feature sets to generate a low-dimensional embedding rich in cross-modal information.

- **Weighted feature reconstruction:** features are dynamically adjusted so that important features are more prominent and unimportant features are suppressed. Here for a single sample, let its visual feature vector be $f_{img} \in \mathbb{R}^{2048}$ and structured feature vector be $f_{tab} \in \mathbb{R}^{128}$, the corresponding gating outputs are scalar g_{img} and g_{tab} , which are first applied separately:

$$\tilde{f}_{img} = g_{img} \times f_{img}, \quad \tilde{f}_{tab} = g_{tab} \times f_{tab} \quad (4)$$

This allows for adaptive amplification or suppression of a certain way of information according to the sample, preventing noise modes from affecting the final representation.

- **Fused Feature Normalization:** To avoid dominating the model learning when the value range of a feature is large after fusion, we splice the weighted two

Commented [14]: Rewrite this section more concisely and integrate it with the original 3.4.4.

Commented [15]: This section has been merged with the original Section 3.3.3.

567 vectors by dimension to get $f_{\text{used}} \in \mathbb{R}^{2176}$, and then apply BatchNorm to
568 normalize the spliced vectors:

569
$$f_{\text{norm}} = \text{Batch Norm}(f_{\text{used}}) \quad (5)$$

570 to eliminate differences in the numerical scales of the two-way features and to
571 mitigate cross-modal internal covariate bias.

572 • **Dimensionality Reduction Projection (DRP):** normalized features are first
573 activated by LeakyReLU, then Dropout regularization is done, and finally
574 mapped to the low-dimensional space:

575
$$h = \text{LeakyReLU}(f_{\text{norm}}), h' = \text{Dropout}(h, p=0.2), z = W_{\text{proj}} \cdot h' + b_{\text{proj}} \quad (6)$$

576 where $z \in \mathbb{R}^{128}$ is the final multimodal embedding. the negative slope of
577 LeakyReLU ensures the preservation of negative features, and Dropout provides
578 additional regularization in small sample scenarios.

Commented [16]: Detailed formula steps have been moved to S5 Appendix. Multimodal Feature Fusion and Dimensionality Reduction

579 Our multimodal fusion layer employs just two fully connected layers to
580 progressively compress and integrate features from 2176 to 512 dimensions, then to
581 128 dimensions. This design achieves significant parameter reduction and structural
582 lightweighting compared to complex architectures like multi-head attention. Batch
583 normalization (BatchNorm) maintains numerical scale consistency between visual and
584 structured features, accelerating training convergence. LeakyReLU activation
585 preserves negative feature information to enhance representational capacity, while
586 Dropout (rate=0.2) applies dual regularization to effectively control overfitting in
587 few-shot scenarios. The fusion process preserves deep cross-modal interactions while

ensuring training stability and inference efficiency, enabling MultiModalFewShotNet to reliably learn robust multimodal embeddings for data-scarce clinical applications.

After multimodal fusion and obtaining the low-dimensional embedding vector, we perform L2 normalization to map it onto the unit sphere. This not only eliminates the magnitude differences between sample vectors, allowing Euclidean distance and cosine similarity to be directly compared within the same metric space, but also simplifies the prototype-based few-shot classification process. In Prototypical Networks, class prototypes are obtained by calculating the simple mean of normalized embedded vectors from the support set, and the normalized prototypes also lie on the unit sphere, ensuring consistency and interpretability in distance metrics between classes[38].

3.5. Few-Shot Learning Strategy

3.5.1. Multi-Prototype Construction

In classical Prototypical Networks (PN)[38], each category typically constructs a single prototype vector - a category representation vector obtained by averaging all features of that category in the Support set. However, this single-prototype approach often fails to capture multimodal or subclass feature variations within categories in real-world complex data scenarios, potentially leading to inadequate category discrimination[39]. To address this limitation, we propose a multi-prototype construction method based on KMeans clustering (**Fig 4**), where multiple prototype vectors collectively represent category feature distributions instead of relying on a

Commented [17]: Retain section headings and core logic, but significantly condense body details and move all technical implementation details of “how to do it” to S6 Appendix. Few-Shot Learning Strategy

Revised:

In classical **Prototypical Networks (PN)**, each class is represented by a single prototype computed as the mean of support embeddings, which may be insufficient to capture **intra-class multimodality** and sub-category structure in real-world data [38, 39]. To address this, we adopt a **KMeans-based multi-prototype** scheme: support-set embeddings of the same class are aggregated into multiple prototypes that jointly describe the class distribution (Fig. 4). The clustering step is used **offline** as an unsupervised aid and does **not** interfere with the supervised training objective; it enriches class structure and improves the stability and accuracy of metric-based inference. For categories with extremely few samples, we apply a **mean-based compensation** to maintain a **fixed number** of prototypes per class, ensuring consistent interfaces during inference. Prior studies also report benefits of multi-prototype representations in settings with pronounced intra-class variation—for example, Zhu et al.’s **Multi-Prototype Prototypical Network** achieves strong results in fine-grained recognition—supporting our design choice [40]. **Full construction details are provided in S6.**

single prototype.

In order to create several prototype centers (Multi-Prototypes) for each category, we first extract the feature representations of each category from the support set using the KMeans approach. The KMeans algorithm's objective function is:

$$\arg \min \{C_k\}_{k=1}^K \sum_{k=1}^K \sum_{x \in C_k} \|x - \mu_k\|^2 \quad (7)$$

where C_k is the set of features in the k th cluster, μ_k is the mean vector of features in the cluster, or the corresponding prototype center, and K is the number of prototypes built for each category ($K=3$ in this paper). The KMeans algorithm divides the data into clusters and finds the center of each cluster by summing the squares of the distances from each sample point to the cluster center to which it belongs. This ensures that the samples in the clusters are as close together as possible. It should be noted that the KMeans algorithm serves solely as an offline unsupervised clustering tool to assist in uncovering intra-category diversity. This approach does not interfere with the labeled supervised training process of the main classification task, but rather enhances the robustness and accuracy of distance metric learning and classification by enriching the structural information of prototype centers.

A category may not be able to produce a sufficient number of clustering centers in situations when there are few samples. In these cases, we make up for this by using the average value of the category features to guarantee that each category has a constant number of prototype centers. Recent research has demonstrated that when there are significant differences in features within a category, the multi-prototype

Commented [18]: Move to S6 Appendix. Few-Shot Learning Strategy

approach can significantly improve classification performance when compared to the single prototype approach. Additionally, the Multi-Prototype PrototypicalNetwork, which was proposed by Zhu[40] and others, achieves good results in the fine-grained image recognition task, further supporting the rationality and efficacy of our approach.

3.5.2. Prototype Normalization and Minimum-Distance Selection

After constructing the prototypes, we perform L2 normalization on all prototypes to eliminate the effect of feature scale differences on subsequent distance calculations. For each prototype μ_k , the normalized prototype is represented as follows:

$$\hat{\mu}_k = \frac{\mu_k}{\|\mu_k\|_2} \quad (8)$$

Subsequently, when classifying samples from a Query set, we compute the Euclidean distance between each query sample feature and multiple prototypes of the corresponding category separately, and select the smallest of these distances as the distance between that query sample and that category:

$$d(x_q, C) = \min_{k=1, \dots, K} \|x - \hat{\mu}_k\|_2 \quad (9)$$

where x_q is the features of the query sample, C is the set of prototypes in a category.

This minimum distance strategy can effectively capture the nearest local features within a category, thus improving the accuracy of classification.

Compared to a single-prototype strategy, multi-prototype approaches can more finely capture intra-class multimodal characteristics, with particularly pronounced benefits in medical few-shot scenarios. This aligns with Tang et al.'s Adaptive

Commented [19]: This is a brief overview; detailed information is provided in S6:

Revised:

To eliminate scale differences and ensure a consistent metric space, **L2 normalization** is applied to all prototypes after construction. During classification, a query's similarity to a class is defined as the **minimum Euclidean distance** to that class's prototypes, aligning the query with its **nearest local pattern** within the class and thereby improving recognition accuracy and robustness (Details are available in S6 Appendix. Few-Shot Learning Strategy)

Prototype Learning (APL), which demonstrated higher accuracy and generalization in remote-sensing image classification through a multi-prototype design[41]. **Table 1** summarizes the core differences and advantages between traditional single-prototype methods and our proposed multi-prototype approach.

Method	Number of Prototypes per Class	Intra-class Feature Representation Ability	Computational Complexity	Handling Method for Insufficient Samples
Prototypical Networks [39]	1	Low	Low	None
Multi-Prototype [41]	≥ 2	Medium	Medium	Simple Mean Compensation
Proposed Method	3	High	Medium	KMeans+Mean Compensation
Matching Networks [42]	N/A	Medium	High	Episodic Training Framework, no fine-tuning required
Relation Networks [43]	N/A	High	High	Learn deep nonlinear metrics, no explicit prototypes required
Transductive Propagation Network [44]	Not Applicable	High	High	Label propagation using the entire query set

Table 1. Comparative Analysis of the Proposed Multi-Prototype Method and Conventional Approaches

- Matching Networks employ an attention mechanism to perform pointwise matching between support set examples and query samples. Through episodic training, they enable rapid generalization in few-shot scenarios without

introducing explicit prototypes[21].

- Relation Networks learn a transferable deep metric function (Relation Module) via a two-branch network, outperforming fixed-distance metrics in feature representation and eliminating the need for prototype construction[42].
- Transductive Propagation Network (TPN) applies a graph-propagation strategy during testing to jointly infer labels across support and query sets, effectively mitigating information scarcity in few-shot settings[43].

In contrast, our method retains the efficiency of prototypical networks while compensating for the limitations of single-center representations through multiple prototypes, achieving a balance between performance and computational cost.

3.6. Loss Function Design

To address three key challenges in clinical few-shot multimodal classification—insufficient intra-class diversity, semantic drift of pseudo-samples, and uneven embedding-space distribution—we propose a multi-objective loss fusion framework. First, a multi-prototype classification loss captures the multimodal, multi-peak feature distributions within each class. Second, an Anchor alignment loss enforces semantic consistency between generated pseudo-samples and their true category anchors. Third, a Batch-Hard triplet loss sharpens boundary discrimination between positive and negative sample pairs. Finally, a weighted combination strategy balances these losses, enabling synergistic optimization of classification accuracy and feature discriminability. This multi-loss design both localizes salient sample

Commented [20]: The detailed content of the four sections (3.6.1; 3.6.2; 3.6.3; 3.6.4) has been moved in its entirety to Supporting information S7. The main text retains the introductory paragraph and appropriately supplements the discussion on these losses and the role of the final weighting.

characteristics and robustly expands the negative-class feature space, significantly improving discrimination and generalization in few-shot scenarios.

3.6.1. Multi-prototype Classification Loss

In order to overcome the shortcoming of traditional prototype networks that cannot adequately express the diversity within the categories with only a single center, we generate $K=3$ prototypes for each category by KMeans on the support set $\{\mu_{c,k}\}_{k=1}^3$ and adopt the “minimum distance” criterion in the query stage for classification. Specifically, let the query sample be characterized as x_q , and its distance to category c is defined as:

$$d_c(x_q) = \min_{1 \leq k \leq 3} \|x_q - \mu_{c,k}\|_2 \quad (10)$$

and use negative distances as categorical logits:

$$l_c(x_q) = -d_c(x_q) \quad (11)$$

Weighted cross-entropy was used for the final loss:

$$L_{\text{proto}} = - \sum_{c=1}^C \omega_c 1\{y_q=c\} \left[l_c(x_q) - \log \sum_{j=1}^C e^{l_j(x_q)} \right] \quad (12)$$

where ω_c is set based on category frequency to mitigate positive and negative category imbalances.

In principle, the multi-prototype approach is equivalent to a multi-peak (Mixture) approximation of the class-conditional distribution $p(x|y=c)$ for each category, i.e., each prototype $\mu_{c,k}$ corresponds to a high-density subgroup. Selecting the center of the nearest subpopulation when querying is able to capture the multimodal structure

Commented [21]: To address three key challenges in clinical few-shot multimodal classification—insufficient intra-class diversity, semantic drift of pseudo-samples, and uneven embedding-space distribution—we propose a **multi-objective loss fusion framework**. First, a *multi-prototype classification loss* captures the multimodal and multi-peak distributions within each class, effectively representing intra-class heterogeneity beyond traditional single-center prototype networks [40,45]. Second, an *anchor alignment loss* ensures semantic consistency between generated pseudo-samples and their true category anchors (e.g., SwAV-derived vectors), thereby mitigating semantic drift and enhancing pseudo-sample reliability [46,47]. Third, a *batch-hard triplet loss* focuses on the most discriminative positive–negative sample pairs, tightening category boundaries and accelerating convergence while improving inter-class separability [48,49]. Finally, these sub-losses are **weighted and combined** into a unified training objective following multi-objective optimization principles [50], which balances classification accuracy, semantic alignment, and embedding-space discriminability. Detailed mathematical formulations, parameter configurations, and ablation analyses of each loss component are provided in **Supporting information (S7 Appendix. Loss Function Design)**

within the category, which in turn forms a segmented decision boundary consisting of Voronoi regions; at the same time, using the minimum distance instead of the mean distance effectively reduces the sensitivity to outliers and noise.

3.6.2. Anchor Alignment Loss

Although AnchorFeatureGenerator is capable of generating diverse pseudo-samples, if its features are semantically offset from the true category center (Anchor), it introduces noise and weakens subsequent prototype learning. For this reason, we introduce an Anchor alignment loss in training by constraining the cosine similarity of each pseudo-sample feature f_i to its corresponding SwAV pre-trained Anchor vector a_{y_i} :

$$L_{\text{align}} = 1 - \frac{1}{N} \sum_{i=1}^N \frac{f_i \cdot a_{y_i}}{\|f_i\|_2 \|a_{y_i}\|_2} \quad (13)$$

where N is the total number of pseudo-samples and label y_i corresponds to the center of the category in which they should be clustered. This alignment strategy imposes semantic consistency constraints on the generated samples directly in the feature space, which significantly reduces the drift between pseudo-samples and the true distribution and improves the validity of pseudo-samples in subsequent prototype construction[46][47]. Compared to adversarial domain-alignment methods, our approach is computationally simpler, integrates seamlessly with multi-prototype classification, and directly preserves angular feature relationships via cosine-similarity optimization, thereby enhancing the discriminative power of the

embeddings.

3.6.3. Batch-Hard Triplet Loss

To further tighten the category boundaries of the feature space, we use the "Batch-Hard" triad loss, which automatically picks only the most discriminative positive/negative pairings of samples inside each mini_batch for optimization. First, we calculate the Euclidean distance matrix ($D_{ij}=\|e_i-e_j\|_2$) between all embedding vectors inside a batch, and then for each sample, we select the "hardest positive sample (d_i^{pos})" with the biggest distance among the same-labeled samples i , and the "hardest negative sample (d_i^{neg})" and:

$$L_{triplet} = \frac{1}{B} \sum_{i=1}^B \max(d_i^{pos} - d_i^{neg} + m, 0) \quad (14)$$

Optimization (bounded hyperparameter, $m=0.5$) Compared to random sampling, Batch-Hard focuses only on the most difficult pairs of samples, which not only avoids the redundant gradient caused by easy-to-classify samples, but also significantly accelerates model convergence and strengthens discriminative ability in the case of fewer samples, particularly when negative samples are scarce and unevenly distributed[48][49]. This module, along with multi-prototype categorization and Anchor alignment loss, creates a more robust and expressive embedding space.

3.6.4. Weighted Combination of Overall Loss

Finally, we weighted and combined the three sub-losses according to the following formula to form the ultimate training objective:

$$L = L_{proto} + \alpha L_{align} + \beta L_{triplet} \quad (15)$$

The weighting design ($\alpha=0.2$, $\beta=0.5$) ensures the multi-prototype classification loss dominates the main task while balancing the constraints of the Anchor alignment loss on pseudo-sample semantic consistency and the Batch-Hard triplet loss's reinforcement of embedding space boundaries through smaller weights. This scheme references Kim et al.'s weight allocation method for sample fusion in class-imbalanced tasks while incorporating Li et al.'s loss balancing experience from tabular data augmentation scenarios[49][50]. Through this synergistic multi-objective loss optimization, the current framework achieves accurate classification decisions in small-sample multimodal scenarios while maintaining clear feature space organization, providing robust support for subsequent model generalization and robustness.

3.7. Training Procedure

The overall training workflow (**Fig 5**) is as follows: first, raw data are standardized and negative-class samples are augmented; then, support and query sets are generated using few-shot episodic sampling. The multimodal feature extraction module separately processes imaging data with ResNet-50 and tabular data with an MLP, and gated fusion produces 128-dimensional embeddings. K-means clustering generates multiple prototypes per class, and three loss terms are computed and jointly optimized. Model parameters are iteratively updated using the Adam optimizer, and the final multimodal fusion model weights—capable of handling both image and clinical inputs—are saved. The workflow incorporates dynamic class weighting and clinical-range validation to ensure reliability and clinical plausibility in data-scarce

scenarios.

4. Model Evaluation

To evaluate our proposed multimodal model, we employed a nested cross-validation strategy with five-fold stratified partitioning to ensure balanced class distributions in each fold. **Fig 6** illustrates the workflow of our evaluation process. This approach was chosen to mitigate bias arising from data imbalance and to ensure robust performance estimates. In each fold, we utilized pretrained submodels—including a multilayer perceptron (MLP) for tabular data, a convolutional neural network (CNN) for imaging data, and our prototype network—to extract features from both modalities. The predictions from these models were then averaged via an ensemble method. Binary classification decisions were made using a global threshold determined by the Youden index to further optimize performance. Model efficacy was quantified using multiple metrics—accuracy, balanced accuracy, precision, recall, F1 score, ROC AUC, and Matthews correlation coefficient (MCC)—computed for each fold and summarized by their mean values.

5. Results

Through systematic multimodal data integration, the fusion-based postoperative AKI prediction model proposed in this study demonstrated significantly greater performance improvements compared to single-modality approaches. The imaging-only model (ResNet50) achieved an AUC of 0.510 in independent validation, while the clinical data model (incorporating preoperative creatinine, lactate, age, etc.)

reached an AUC of 0.578. Although both models exhibited some predictive capability, they showed clear limitations. By innovatively combining these two modalities with prototype learning, the final model achieved substantially enhanced performance (AUC=0.816).

Through SHAP-based interpretability analysis, this study effectively reveals the clinical feature contributions underlying the model's decision-making process, enabling clinicians to intuitively understand the biological basis of predictions and increasing their trust in the model. This transparent interpretation mechanism allows physicians to both comprehend general decision patterns and validate prediction rationality for individual cases. Consequently, it significantly enhances the efficiency and reliability of preoperative risk assessment in human-AI collaboration, facilitating clinical implementation of the AI model.

Although the representativeness of our sample and the generalizability of derived thresholds require further improvement, our multimodal fusion framework has demonstrated significant clinical value. In future work, multi-center, large-scale cohorts encompassing diverse surgical procedures will expand sample coverage and enable more granular subgroup analyses. Additionally, implementing dynamic thresholding strategies tailored to individual patient characteristics and clinical contexts will further enhance personalized adaptability. Concurrent integration of intraoperative real-time physiological monitoring data and richer imaging and histological features is expected to establish a closed-loop system uniting preoperative

risk assessment with intraoperative early warning. These advances will accelerate the translation of our model from research validation to routine clinical application, providing intelligent support for precise perioperative decision-making.

5.1. Model Performance

Using five-fold cross-validation, the postoperative acute kidney injury (AKI) prediction model developed in this study demonstrated robust predictive performance. The model exhibited stable prediction across different data splits, with an average AUC of 0.816 ± 0.036 . The overall ROC curve (**Fig 7**) remained consistently above the random classification baseline, while individual fold AUC values ranged from 0.764 to 0.863, with the fourth fold achieving optimal performance (AUC=0.863, accuracy=0.811). The model showed excellent sensitivity (recall 0.788 ± 0.074), enabling effective identification of AKI-positive cases and reducing underdiagnosis risk in clinical practice. However, precision (0.729 ± 0.038) exhibited greater variability, particularly in fold 1 where it decreased to 0.679, indicating the need for further optimization to reduce false positives in certain data subsets. Detailed performance metrics are shown in the per-fold confusion matrices (**Fig 8**) and ROC curves (**Fig 9**).

The optimal threshold determined by Youden's index was 0.332, which demonstrated good performance in balancing sensitivity and specificity. However, its applicability requires careful consideration in specific populations or clinical contexts. Future studies should explore threshold adjustments tailored to particular medical

scenarios (e.g., high-risk surgical patients or elderly populations) to further optimize predictive performance across different applications. From a clinical decision-making perspective, comprehensive analysis of ROC curves and confusion matrices supports the model's potential value for preoperative risk stratification and postoperative intervention planning. Additional research should focus on external validation and real-world clinical evaluation to enhance the model's generalizability and practical utility.

5.2. SHAP-based Feature Importance

In this study, the SHAP (SHapley Additive exPlanations) method was applied to evaluate model interpretability and analyze the contribution of each preoperative clinical variable to AKI prediction (**Fig 10**). SHAP values quantify the average influence of individual features on model outputs, revealing potential biological mechanisms behind the model's decisions. Ranked by global mean absolute SHAP values:

- **Preoperative serum creatinine:** the most predictive indicator, reflecting the baseline renal function's decisive role in postoperative AKI risk and underscoring the key clinical utility of traditional renal markers in early warning.
- **Age:** increased susceptibility to AKI in elderly patients highlights the need to consider physiological aging in preoperative assessments.
- **Preoperative lactate:** signifies the important influence of perioperative

tissue hypoperfusion and metabolic disturbances on AKI pathogenesis.

Moreover, routine preoperative biochemical markers such as AST and blood urea nitrogen contributed moderately to prediction, whereas chronic comorbidities—hypertension, diabetes mellitus, and solitary kidney status—showed lower SHAP values, likely reflecting the small number of solitary kidney cases in our cohort and the model’s limited learning of this feature’s potential impact.

Overall, SHAP analysis not only confirmed the model’s effective capture of key clinical predictors but also provided clear targets for clinical intervention: preoperative monitoring and optimization of creatinine, lactate, and age-related risk factors are recommended to support precise risk stratification and the development of effective preventive strategies to reduce postoperative AKI incidence.

6. Discussion

This study’s postoperative AKI prediction algorithm offers significant methodological innovation. Traditional deep learning techniques perform poorly on small-sample data because of the short data samples, particularly when dealing with unbalanced data that is prone to overfitting. Because of this, despite having little data, Few-Shot Learning (FSL), a crucial area of study in artificial intelligence, may learn efficiently and produce precise predictions. This study successfully avoids the overfitting issue in the conventional methods and effectively enhances the model’s generalization capacity by incorporating small-sample learning and data enhancement strategies, which guarantees the model’s strong performance under restricted

870 samples. The model maintains clinical value while achieving excellent predicted
871 accuracy through the integration of imaging data with routine preoperative clinical
872 markers. The model's operability in the clinical setting is further enhanced by the fact
873 that, according to systematic analysis, it only has to include 11 standard preoperative
874 examination indicators to attain good predictive performance. In the meantime, the
875 use of SHAP interpretative analysis guarantees the interpretability of model outcomes
876 and successfully improves the transparency of model decision-making.

877 Our study creatively added the clinical feature of "sole kidney or not" as an
878 independent component in the selection of characteristics, in addition to combining
879 the imaging and tabular data. We believe that the potential contribution of this feature
880 to the model should not be overlooked, even though the SHAP analysis revealed a
881 small effect of this factor on the model. This may be related to the small sample size
882 of patients with solitary kidneys (only 7 out of 194 cases), and that this feature is
883 based not only on clinical data (such as the patient's history) but also may be
884 associated with imaging findings (such as the kidney's structural features), providing a
885 multidimensional perspective that is especially relevant in certain high-risk patient
886 populations. The solitary kidney factor can become increasingly significant in the
887 model as the sample size grows. Additionally, imaging data was crucial to our
888 investigation. The current preoperative indicators can already be used for effective
889 prediction, even though detailed imaging indicators like tumor size and location were
890 left out. Future optimization of the imaging feature selection process can also be

carried out, and additional structural and functional imaging data can be integrated to enhance the model's accuracy and clinical applicability.

Furthermore, the tabular data selection process adhered to the current international standards, which include diabetes[51] and hypertension[52] as predictors. According to earlier research, the two main chronic conditions influencing renal function are diabetes mellitus and hypertension, both of which have long-term impacts on the kidney through chronic pathways[53][54]. Consequently, a more thorough clinical viewpoint for forecasting postoperative AKI is offered by the model's incorporation of preoperative blood pressure and glucose levels, as well as the presence of diabetes and hypertension.

From a pathophysiological point of view, the model's primary predictors not only support current clinical knowledge but also offer new insights into the possible mechanisms through which preoperative metabolic status (such as lactate level) influences the development of postoperative AKI. With regard to clinical translation, the current model differs from earlier prediction systems that depend on specialized testing methods; it uses standard preoperative screening signs, which increases its generalizability. Specifically, the model's probabilistic risk score values can be easily incorporated into current clinical decision pathways, significantly aiding physicians in making preoperative decisions.

Nonetheless, there are still certain restrictions on the current study. In addition, the current model only provides static preoperative prediction and excludes

912 intraoperative and postoperative dynamic monitoring data, which somewhat affects
913 the real-time and accuracy of the prediction. The single-center data source limits the
914 model's generalization ability, particularly the validity in special populations (such as
915 children, patients with a single kidney, or patients with severe renal impairment),
916 which has not yet been confirmed. Developing dynamic prediction models to
917 maximize prediction ability in minority populations and investigating deep integration
918 with clinical information systems and hospital electronic medical records to achieve
919 real-time dynamic monitoring and management of postoperative AKI risk should be
920 the main goals of future research.

921 The model's greatest usefulness from a clinical application standpoint is seen in
922 its importance for preventative medicine. Clinicians can successfully lower the
923 incidence of postoperative AKI by implementing measures for high-risk patients
924 beforehand, such as limiting the use of nephrotoxic drugs and optimizing volume
925 control techniques, through accurate preoperative risk assessment. In addition to being
926 a significant avenue for perioperative management advancement, our prediction-based
927 medical model offers a crucial theoretical and practical foundation for the future
928 development of an intelligent postoperative complication warning and intervention
929 system.

930 7. Acknowledgments

931 Special thanks to Associate Professor Jun Li for his invaluable guidance, which
932 made this project possible. I am also grateful to Attending Physician Zhen Yang for

Commented [22]: (Lines 901–909) Claims the study possesses significant innovation and clinical impact, but fails to provide sufficient justification. (Strongly asserts novelty and clinical impact, yet lacks adequate evidence. Should include references to early prediction systems and adjust or provide data supporting statements about physician assistance.)

Revised 901–905:

Future research should focus on developing **dynamic and continuously updated prediction models** to enhance robustness and generalizability across diverse surgical populations. In addition, establishing **temporal and multicenter validation frameworks** will be crucial to

Commented [23]: Add a section after this description of “limitations”

“(Temporal Validity)” discussion:

Add:

Additionally, the **temporal validity** of the model should be acknowledged. As the model was developed from historical single-center data, future changes in perioperative workflows, surgical techniques and instruments, anesthesia and hemodynamic management, and laboratory testing methods may alter the distribution of input features and the incidence of AKI, potentially leading to **temporal performance drift**. To mitigate this risk, we plan to perform **temporal external and multicenter validation**, and to establish **regular performance monitoring and recalibration procedures**.

Commented [24]: From a clinical perspective, the **potential usefulness** of our model lies in its contribution to **perioperative preventive medicine**. By accurately identifying high-risk patients before surgery, clinicians may implement **targeted preventive strategies**—such as minimizing nephrotoxic drug exposure, maintaining hemodynamic stability, and optimizing volume management—to **potentially reduce postoperative AKI incidence** 【1,2】.

Furthermore, building upon advances in **early AKI warning systems and EMR-based decision-support tools** 【3–5】, integration of our **surgery-specific model** into these platforms could facilitate **real-time risk monitoring and proactive intervention**, providing a **conceptual and**

his insightful comments on the manuscript. Additionally, I extend my appreciation to all colleagues in the lab for their unwavering support and collaboration—this research would not have been possible without your contributions.

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Figure information

Fig 1. Participant Eligibility Criteria. Participant screening and grouping process.

Researchers screened individuals according to predefined inclusion criteria, yielding a final study cohort of 194 subjects.

Fig 2. Three-step data augmentation strategy for negative-class samples. first, diverse negative pseudo-samples are generated using Anchor vectors; second, Euclidean distances among negative samples are computed to identify nearest neighbors for interpolation, enhancing local diversity; finally, the pseudo-samples are combined with real negative samples.

Fig 3. Multimodal Feature Fusion Network.

Fig 4. Process for Generating Multiple Prototypes. Samples are first grouped by class, then K-means clustering is used to compute centroids for each class; if a class has insufficient samples, mean imputation is applied; finally, the centroids are L2-normalized to produce the set of multiple prototypes.

Fig 5. Training Workflow.

Fig 6. Model Evaluation Workflow.

Fig 7. Overall ROC Curve

Fig 8. Five-fold Cross-validation – Confusion Matrix

- 1101 **Fig 9. Five-fold Cross-validation – ROC Curves**
- 1102 **Fig 10. Feature Importance for Postoperative AKI Prediction**
- 1103 **Table 1. Comparative Analysis of the Proposed Multi-Prototype Method and**
- 1104 **Conventional Approaches**

Manuscript title: Few-Shot Prediction of Post-Fusion Acute Kidney Injury Using Multimodal Radiomics-Clinical Feature-Based Prototype Networks

Journal: PLOS ONE

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Dear Dr. Irfan Ullah and Reviewers,

We sincerely thank you for the constructive feedback. We have revised the manuscript accordingly. Below, we provide a point-by-point response—each item begins with a bold Comment label followed by our **Response**—and we indicate the exact locations of the corresponding changes in the **Revised Manuscript** (with Track Changes) and the **Clean Manuscript** (page and line numbers). For substantial rewrites, replacement text is provided in the **Appendix**.

Reviewer #1 — 1. Page 12, Lines 251–258 (Manuscript length & repetition; KDIGO definition duplicated)

Comment: “The manuscript is overly long and contains considerable repetition with unclear structure. For example, the KDIGO definition of AKI appears verbatim in both the Introduction and Methodology. Please streamline the text, remove redundancies, and focus on the most relevant content.”

Response: Thank you for this important comment. We have substantially streamlined and restructured the relevant sections and have removed the verbatim KDIGO definition from this (Methodology) passage. The formal KDIGO definition now appears only once in the Introduction with citation. In the Methodology, we no longer repeat the definition; instead, we include a one-sentence cross-reference to the Introduction and retain only the baseline SCr selection rule necessary for outcome adjudication, thereby avoiding redundancy while ensuring reproducibility and auditability.

Actions taken:

- **Remove the duplicate definition:** Deleted the verbatim KDIGO definition from Methodology; retained a single, authoritative KDIGO definition only in the Introduction (with references).
- **Retain essential rule and add cross-reference:** In Methodology (Outcome definition), added a one-sentence note—“AKI definition, see Introduction”—and kept only the baseline SCr selection rule to support outcome adjudication; KDIGO thresholds and time windows are not repeated.
- **Consolidate redundant statements:** Merged statements regarding “retrospective design, anonymization, and lack of access to identifiable information” into a single paragraph (see replacement text below), and centralized ethics and data-availability details in their dedicated sections/Supplement to avoid scatter.

Location in the revised manuscript:(Manuscript)

- Introduction—Background/Definitions (the KDIGO formal definition appears only once): **page 3, lines 45-48.**

- Study population / Data sources (revised paragraph replacing the prior KDIGO duplication): **page 12, lines 251–258.**

Reviewer #1 — 2. Page 6, Lines 115–131 (Rationale for focusing on AKI after laparoscopic renal surgery)

Comment: “The rationale for focusing on AKI after laparoscopic renal surgery (partial resection/nephrectomy) is underdeveloped. The manuscript does not explain why this surgical context was chosen or why a dedicated prediction model is needed. Please clarify in the Introduction why this setting is a priority for investigation.”

Response: We appreciate this important suggestion and agree that the original manuscript did not sufficiently articulate the study rationale. In accordance with your advice, we have substantively rewritten and expanded the relevant paragraph in the Introduction to:

- (1) explicitly state that laparoscopic nephrectomy involves dual renal stress mechanisms—pneumoperitoneum-induced reductions in renal perfusion and ischemia–reperfusion injury during hilar clamping—thereby explaining, on a pathophysiological basis, why this setting is more prone to AKI;
- (2) add the incidence range (15%–25%) under KDIGO criteria and the robust association between postoperative AKI and long-term renal function decline/new-onset CKD, underscoring clinical importance;
- (3) clarify that generic perioperative AKI models, mostly derived from cardiac or heterogeneous surgical cohorts, omit procedure-specific variables (e.g., pneumoperitoneum pressure, warm ischemia time), limiting transportability and predictive performance in this setting;
- (4) consequently, highlight the need for a surgery-specific prediction model tailored to this context to address the research gap and clinical demand. All assertions are supported with targeted references.

Actions taken:

- Replaced the original lines 115–118 with a structured rationale covering four

layers—mechanism, epidemiology, prognosis, and model gap—and added/updated references to support each key point.

- Ensured consistency by introducing and defining procedure-specific variables (pneumoperitoneum pressure, warm ischemia time) in the subsequent Methods and variable lists, thereby closing the loop with the Introduction.

Location in the revised manuscript: (Manuscript) Introduction, page 6, lines 115–131

Reviewer #1 — 3. Pages 18–44 (Methods overly long; details should be moved to Supplement)

Comment: “The Methods section is excessively long (approximately 20 pages) and is not appropriate for the main manuscript. Only essential methodological details should be retained, while detailed definitions and extended technical explanations should be moved to the Supplement to improve readability and focus.”

Response: We appreciate this key suggestion. We agree that the main-text Methods should focus on core workflow and decisions, while implementation details, parameters, and extended experiments should be placed in the Supplement. Accordingly, we performed a structural condensation and rewrite of the Methods, retaining only the key elements needed for reproducibility in the main text with pointer citations, and moving all additional technical details to the Supplement.

1. Modification 1 (page 18, lines 369–392)

Actions taken:

Merge and refocus: We merged the original 3.2 Data Preprocessing and 3.3 Data Augmentation into 3.2 Data Preprocessing and Augmentation, retaining only:

- Clinical branch: median imputation and Z-score normalization ;
- Imaging branch: YOLO-based ROI cropping and uniform resolution (with a pointer to Supporting information S2);
- Class imbalance: feature-space interpolation & weighted sampling &

class-weighted loss within the multimodal framework;

- Anchor-based pseudo-sample strategy presented as a high-level overview (linked to Fig. 2), with algorithmic details moved to Supporting information S2.

Details moved to the Supporting information:

- Supporting information S1 (Data Imputation and Standardization): implementation details for imputation/standardization and train–inference consistency;
- Supporting information S2 (Data Augmentation, Class Balancing, and Feature Extraction): augmentation parameters (e.g., $\pm 5^\circ$ rotation, $\pm 5\%$ brightness/contrast, 224×224), backbone and feature dimensions (e.g., ResNet-50 without classification head, 2048-D), SMOTE-inspired interpolation, WeightedRandomSampler & class-weighted loss, and the three-step anchor-based generation with thresholding.

Remove redundancy and engineering minutiae: Eliminated repeated descriptions of standardization, duplicate phrasing, library/function names, and serialization steps from the main text; replaced with concise pointers “see Supporting information S1/S2.”

Length reduction: The main-text Methods are substantially shorter than in the original submission (redundancies and implementation details removed), yielding a tighter, clearer presentation that highlights study design and primary innovations.

Location in the revised manuscript:

Methodology — 3.2 Data Preprocessing and Augmentation, **page 18, lines 369–392**; Fig. 2 (anchor-based strategy schematic).

Supporting information S1: Data Imputation and Standardization; Supporting information S2: Data Augmentation, Class Balancing, and Feature Extraction.

2. Modification 2 (page 19, lines 393–408)

Actions taken:

- **Pruned & merged:** Deleted the former 3.4.1 Design Rationale and Architecture

and merged its key points into the opening paragraph of the fusion section (now 3.3).

- **Retained essentials only:** Kept a high-level description of the architecture and key steps to avoid overlap with figures/other subsections.

Location in the revised manuscript:

Methodology — 3.3 Multimodal Feature Fusion Network (one-paragraph summary after merging), **page 19, lines 393–408.**

3. Modification 3 (page 20, lines 410–418)

Actions taken:

- **Pruned and merged:** Deleted content in the original 3.4.2 that duplicated the overview (e.g., background suppression from ROI cropping, 2048-D GAP outputs, flexible fine-tuning, and inference efficiency) and rewrote the subsection as a concise **3.3.1 Feature Extraction Module**.
- **Retained essential points only:** In the main text, we keep two high-level items: **Image feature encoder:** pretrained ResNet-50 with YOLO-based ROI cropping to obtain high-level visual features; **Tabular feature encoder:** shallow MLP + ReLU + Dropout to produce a compact structured representation; and add a one-sentence pointer to the Supporting information for implementation details.
- **Moved details to the Supporting information:** Transferred network architecture, parameter settings, and training strategy to Supporting information S3 (S3.1–S3.3), including ResNet-50 without the classification head + GAP (2048-D), ROI pixelization and resizing to 224×224, a two-layer MLP (ReLU / Dropout = 0.2), and output dimensions (image 2048, tabular 128).

Location in the revised manuscript:

Methodology — **3.3.1 Feature Extraction Module, page 20, lines 407–418** (concise description with a terminal pointer to Supporting information S3).

Supporting information S3 (S3.1 Image Feature Encoder; S3.2 Tabular Feature Encoder; S3.3 Module Parameters and Compatibility).

4. Modification 4 (page 20, lines 419–434)

We merged the former 3.4.3 (Dynamic Gated Fusion Mechanism) and 3.4.4 (Multimodal Feature Fusion and Dimensionality Reduction) into the new 3.3.2 Dynamic Gated Fusion and Dimensionality Reduction. In the main text, we now provide a single, high-level paragraph describing the fusion logic and dimensionality-reduction flow, while formulas and parameter details are moved to the Supporting information, improving readability and focus.

Actions taken:

- **Pruned & merged:** Removed repetitive/stepwise exposition across the two subsections and integrated the pipeline—sample-wise sigmoid gating → weighted reconstruction → lightweight two-layer projection to 128-D → L2 normalization → prototype mean computation—into a coherent description under the new 3.3.2.
- **Essential information retained:** We state that sample-wise sigmoid gating replaces softmax competition to jointly amplify informative modalities and down-weight unreliable inputs; the gated vector is projected via two lightweight layers to a 128-D embedding, then L2-normalized to the unit sphere to ensure metric consistency and interpretability with Prototypical Networks.
- **Details moved to the Supplement:** Formal derivations, network parameters, and implementation details are now in Supporting information S4 (Dynamic Gated Fusion Mechanism) and Supporting information S5 (Multimodal Feature Fusion and Dimensionality Reduction), avoiding length in the main text.
- **No extra hyperparameter tuning:** We clarify that this design improves sensitivity and generalization under small-sample and class-imbalanced conditions without introducing additional hyperparameter tuning.

Location in the revised manuscript:

Methodology — 3.3.2 Dynamic Gated Fusion and Dimensionality Reduction, pages 20–21, lines 419–434.

Supporting information S4 (Dynamic Gated Fusion Mechanism); Supporting

information S5 (Multimodal Feature Fusion and Dimensionality Reduction).

5. Modification 5 (page 21, lines 436–452)

Actions taken:

- Revised and condensed the original 3.5.1 Multi-Prototype Construction and moved implementation details to Supporting information S6 (now presented as 3.4.1 Multi-Prototype Construction in the main text).
- **Key points retained** : We note that a single prototype is insufficient when there is intra-class multimodality/sub-category structure; we therefore adopt a KMeans-based multi-prototype scheme on support-set embeddings, where multiple prototypes jointly describe each class distribution. For classes with very few samples, we apply a mean-based compensation to maintain a fixed number of prototypes. We also cite prior work showing the benefits of multi-prototype representations in fine-grained / high intra-class variability settings.
- **Details moved to the Supplement:** The clustering objective, choice of number of prototypes, and implementation/training interface are all placed in Supporting information S6 (including formulas and implementation notes in S6.1). The main text points readers to “see Supporting information S6” and omits derivations and parameters.
- **Tightened exposition:** The main text emphasizes that this clustering step is an offline, unsupervised aid that does not interfere with the supervised training objective, while enriching class structure and improving the stability and accuracy of metric-based inference.

Location in the revised manuscript:

Methodology — 3.4.1 Multi-Prototype Construction, page 21, lines 436–452.

Supporting information S6 (Few-Shot Learning Strategy), S6.1 Multi-Prototype Construction Details (including formulas and implementation details).

6. Modification 6 (page 22, lines 453–466)

Actions taken:

- **Core logic retained:** We state that all prototypes are L2-normalized to eliminate scale differences. During classification, similarity to a class is computed as the minimum Euclidean distance to that class’s prototypes, aligning each query with the nearest local pattern within the class and thereby improving accuracy and robustness.

Location in the revised manuscript:

Methodology — 3.4.2 Prototype Normalization and Minimum-Distance Selection, page 21, lines 453–466.

Supporting information S6.2 (Prototype Normalization and Minimum-Distance Selection; formulas and implementation details).

7. Modification 7 (page 24, lines 482–501)

Actions taken:

- **Moved details to the Supplement:** We transferred the detailed derivations, parameter settings, and ablation studies from the original 3.6.1–3.6.4 to Supporting information S7. In the main text, we now provide a single-paragraph overview under 3.5 Loss Function Design, clarifying the roles and rationale of the three losses and their final weighting to improve readability and focus.
- **Overview retained in the main text (functions and roles):** 3.5 Loss Function Design—**Multi-prototype classification loss:** captures multi-peak/multimodal intra-class structure, compensating for the limitations of single-center prototypical networks; **Anchor alignment loss:** enforces semantic consistency between pseudo-samples and class anchors (e.g., SwAV vectors), suppressing semantic drift and improving pseudo-sample reliability; **Batch-hard triplet loss:** focuses on the most discriminative positive–negative pairs, tightening inter-class boundaries, accelerating convergence, and improving separability; **Weighted fusion:** combines the sub-losses into a unified objective to balance accuracy,

semantic alignment, and embedding-space discriminability.

- **Details deferred to the Supplement:** Mathematical formulations, parameter configurations, and ablation analyses for each loss are presented in S7. The main text references these details via “see Supporting information S7” and does not repeat derivations or engineering specifics.
- **Streamlining and consistency:** We removed technical statements that duplicated other sections, and unified terminology and citation ranges. Loss weights and sensitivity analyses are reported in S7, aligned with the Results discussion.

Location in the revised manuscript:

Methodology — 3.5 Loss Function Design, page 24, lines 482–501.

Supporting information 7 (formulas, parameter settings, and ablation studies for all losses).

Reviewer #1 — 4. Page33 lines 669-678. (Temporal validity not discussed)

Comment: “Temporal validity should be explicitly discussed; the model needs periodic retraining and external validation on newer data, otherwise performance may degrade.”

Response: Thank you for highlighting this important limitation. We have added a new “Temporal Validity” paragraph to the Discussion—Limitations section. We explicitly acknowledge that a model trained on historical single-center data may experience performance drift over time. We also outline mitigation strategies, including temporal external and multicenter validation, routine performance monitoring and recalibration, and integration with hospital information systems to ensure stability and applicability in real-world clinical settings. In addition, we describe future work on dynamic, continuously updated prediction frameworks and clarify current limitations in generalizing to specific populations (e.g., patients with a single kidney, severe renal impairment, or children), together with planned improvements.

Actions taken:

- Added a dedicated subsection after Limitations discussing temporal validity and the risk of performance degradation as clinical practice evolves and data distributions shift.
- Committed to temporal external + multicenter validation and to establishing ongoing monitoring and recalibration procedures.
- Emphasized integration with hospital information systems to achieve temporal stability and usability in real-world deployment.
- Outlined future directions for dynamic/continually updated models and incorporation of intraoperative/postoperative dynamic data.

Location in the revised manuscript: (Manuscript)

Discussion — Limitations (new “Temporal Validity” paragraph), **pages 33, lines 669-678.**

Reviewer #1 — 5. Lines 901-909 (Claims of novelty and clinical impact are too strong and insufficiently supported)- page33, lines 664-668.

Comment: “The statements about the study’s novelty and clinical impact are overly strong without sufficient evidence. Please add citations to early prediction/early warning systems and temper—or provide data to support—claims about assisting physicians.”

Response: Thank you for this important comment. We have tempered the language, added background citations, and clarified the scope:

1. In the Discussion, we revised statements about “clinical impact/assisting physicians” to a cautious, hypothesis-generating framing, avoiding definitive claims beyond the presented evidence.
2. We softened and forward-looking-framed remarks on “clinical integration,” emphasizing that prospective evaluation, workflow-integration studies, and external validation are required before routine use.
3. We focused future work on dynamic, continuously updated prediction

frameworks and on temporal and multicenter validation to ensure reliability across time and institutions (see the replacement text below).

Actions taken:

- **Tone down & clarify boundaries:** Deleted/rephrased overly strong language; changed phrases like “assist physicians” to potential/conditional wording.
- **Add citations:** Inserted representative references on early prediction/early warning systems and perioperative AKI risk tools in the relevant Discussion paragraph(s) to support context and comparisons.
- **Prospective roadmap & validation:** Added forward-looking statements on dynamic models, temporal/multicenter validation, and information-system integration (see replacement text).

Location in the revised manuscript: Discussion — Clinical Implications/Future Work, page33, lines 664–668.

Reviewer #1 — 6. (Overstated claims about novelty and clinical impact)

Comment: The wording on “clinical impact/assisting physicians” should be tempered, and citations to early warning/EMR decision-support systems should be added to provide adequate context and support.

Response: Thank you for this important suggestion. We have toned down the language and strengthened the evidence base as follows:

- Reframed statements about “clinical impact/assisting physicians” as potential contributions and hypothesis-generating, avoiding definitive claims beyond the current evidence;
- Added targeted evidence linking perioperative preventive measures (e.g., minimizing nephrotoxic exposure, maintaining hemodynamic stability, optimizing volume management) to AKI risk reduction;
- Incorporated representative studies on early AKI warning systems and EMR-based decision-support to situate our work within the existing literature;

- Positioned our model as a surgery-specific risk identification module with potential for integration into existing platforms, while emphasizing that prospective evaluation and workflow-integration studies are prerequisites for any routine clinical use.

Actions taken:

- Replaced the original paragraph with a version that uses tempered wording and additional citations;
- Added conditional, forward-looking phrasing (e.g., “may/has the potential to”) in adjacent sentences, and pointed to the need for external/temporal and multicenter validation as the next steps.

Location in the revised manuscript: (Manuscript)

Discussion — Clinical Implications/Future Work, **page 33, lines 679–688.**

Reviewer #1 — 7. (Language & formatting: typos, spacing, inconsistent styles)

Comment: Please correct typographical/spacing errors and ensure consistent formatting that complies with PLOS ONE style (templates, file naming, figures/tables, references, headings, and supplementary materials).

Response: Thank you for pointing this out. We conducted a comprehensive language and formatting sweep and aligned the manuscript with PLOS ONE requirements. Key updates are summarized below.